Medium Access Sub Layer
Unit-III

OSI

- Application
- Presentation
- Session
- Transport
- Network
- Data Link
- Physical

LOGICAL LINK sublayer

Framing

Error control

Flow control

MEDIA ACCESS sublayer

Transmission/reception of frames

The Medium Access Sub layer

deals with

BROADCAST NETWORKS AND THEIR PROTOCOLS

*Broadcast channels are sometimes referred to as **multi-access channels** or **random access channels.***

Topics

- Introduction
- Channel Allocation problem
- Multiple Access Protocols
- IEEE Standard 802 for LANs
- Wireless LAN
- Bridges & its types.

Introduction

- *Medium Access Control (MAC) sub layer is part of Data Link layer.*
- *In fact, it is the bottom part of DLL (interfacing with the physical layer)*
- *This chapter deals with broadcast networks*

The Channel Allocation Problem

- *Static Channel Allocation in LANs and MANs*
- *Dynamic Channel Allocation in LANs and MANs*

Channel Allocation problem

- *Static Channel Allocation in LANs and MANs*
- *FDM & TDM*
- <u>FDM:</u>

1. The traditional way of allocating a single channel, among multiple competing users is Frequency division multiplexing(FDM).
2. If there are N users, the bandwidth is divided into N equal-sized portions, each user being assigned one portion.
3. Here each user has a private frequency band, there is no interference between users.

4. When there is only a small and constant number of users, each of which has heavy load of traffic, FDM is simple and efficient.
5. When number of senders is large and continuously varying, FDM presents few problems.
6. If spectrum is cut up into N regions and <N users are interested in communication, the spectrum is wasted.
7. If more than N users want to communicate, some of them will be denied permission for lack of bandwidth.
8. Assuming that the number of users be held constant at N, dividing the channel into sub-channels is inefficient.
9. The basic problem is, when some users are quiet their bandwidth is simply lost. They are not using it, and no one else is allowed to use it either.

Performance of static FDM from a simple queueing theory is as follows:

- Assume that the mean time delay, T, for a channel of capacity C bps, with an arrival rate of λ frames/sec, each frame having a length from an exponential probability density function with $1/\mu$ bits/frame.
- With these, the arrival rate is λ frames/sec and the service rate is μC frames/sec.
- From queueing theory it can be shown that for Poisson arrival and service times,

$$T = \frac{1}{\mu C - \lambda}$$

- Now let us divide the single channel into N independent sub-channels, each with capacity C/N bps. The mean input rate on each sub-channels will be λ/N.
- Recomputing T we get

$$T_{FDM} = \frac{1}{\mu(C/N)-(\lambda/N)}$$

$$= \frac{N}{\mu C - \lambda}$$

$$= NT$$

The mean delay using FDM is N times worse than if all the frames were somehow magically arranged orderly in a big central queue.

TDM:

- The same arguments that apply to FDM also apply to TDM.
- Each user is statically allocated every N^{th} time slot. If a user does not use the allocated slot, it just lies fallow.
- The same holds if we split up the network physically.

Note: None of the traditional static channel allocation methods work well with bursty traffic.

Dynamic Channel Allocation in LANs and MANs

- *Station Model.*
- *Single Channel Assumption.*
- *Collision Assumption.*
- *(a) Continuous Time.*
  (b) Slotted Time.
- *(a) Carrier Sense.(LAN)*
  (b) No Carrier Sense.(SATELLITE)

1.Station Model:

- The model consists of N independent Stations, each with a program or user that generates frames for transmission. Stations are called Terminals.
- The probability of a frame being generated in an interval of length Δt is $\lambda \Delta t$, where λ is a constant(arrival rate of new frames).
- Once a frame is generated the station is blocked and does nothing until the frame has been successfully transmitted.

2. Single Channel Assumption:

- A single channel is available for all communication.
- All stations can transmit on it and all can receive from it.

3. Collision Assumption:

- If two frames are transmitted simultaneously, they overlap in time and the resulting signal is garbled. This event is called a Collision.
- All stations can detect collisions.
- A collided frame must be transmitted again later.

4a. Continuous Time:

- Frame transmission can begin at any instant.
- There is no master clock dividing time into discrete intervals.

4b.Slotted Time:

- Time is divided into slots. Frame transmission always begins at the start of a slot.
- A slot can have 0,1, or more frames, corresponding to an idle slot, a successful transmission, or a collision respectively.

5a. Carrier Sense:

- Stations can tell if the channel is in use before trying to use it.
- If the channel is sensed as a busy, no station will attempt to use it until it goes idle.

5b. No Carrier Sense:

- Stations cannot sense the channel before trying to use it.
- They just go ahead and transmit. Only later can they determine whether the transmission was successful.

Multiple Access Protocols

- *ALOHA*
- *Carrier Sense Multiple Access Protocols*
- *Collision-Free Protocols*
- *Limited-Contention Protocols*
- *Wavelength Division Multiple Access Protocols*
- *Wireless LAN Protocols*

Medium Access Sub Layer

ALOHA

Aloha

- *Pure ALOHA (Mr. Norman Abramson in 1970s)*
- *Slotted ALOHA (Mr.Roberts in 1972)*

ALOHA system used to ground based radio broadcasting.

Pure ALOHA

- *Users transmit whenever they have data to be sent.*

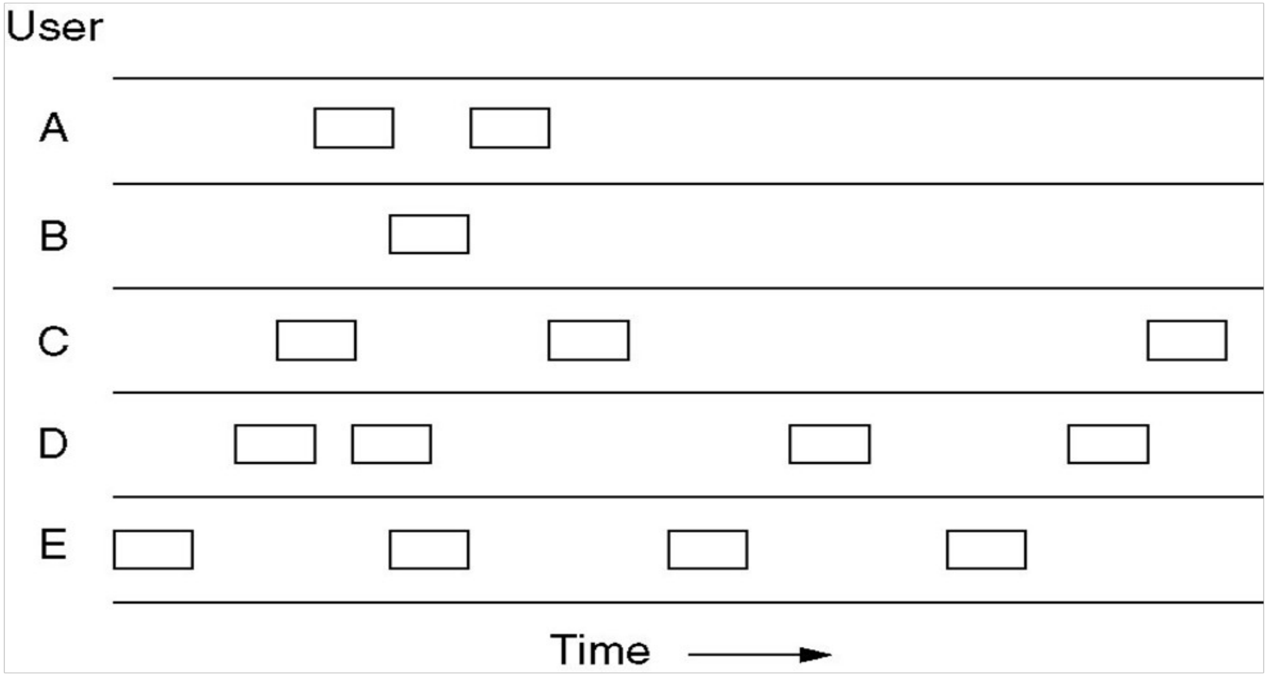


Fig. In pure ALOHA, frames are transmitted at completely arbitrary times

- *Systems in which multiple users share a common channel in a way that can lead to conflicts are widely known as **contention** system.*
- *The frame size is fixed because the throughput of ALOHA systems is maximized.*
- *We need to resend the frames that have been destroyed during transmission.*
- *A collision involves two or more stations. If all these try to resend their frames after the time-out, the frames will collide again.*
- *Pure ALOHA states that when the time-out period passes, each station waits a random amount of time before resending its frame(T_B).*

- *Pure ALOHA has a second method to prevent congesting the channel with retransmitted frames. After a maximum no.of retransmission attempts K_{max}, a station must give up and try later.*
- *The time-out period is equal to maximum possible round-trip propagation delay, which is twice the amount of the time required to send a frame between the two most widely separated stations($2 * T_p$).*
- *The back-off time TB is a random value that depends on K.*

  (binary exponential back-off)

Figure 12.4 *Procedure for pure ALOHA protocol*

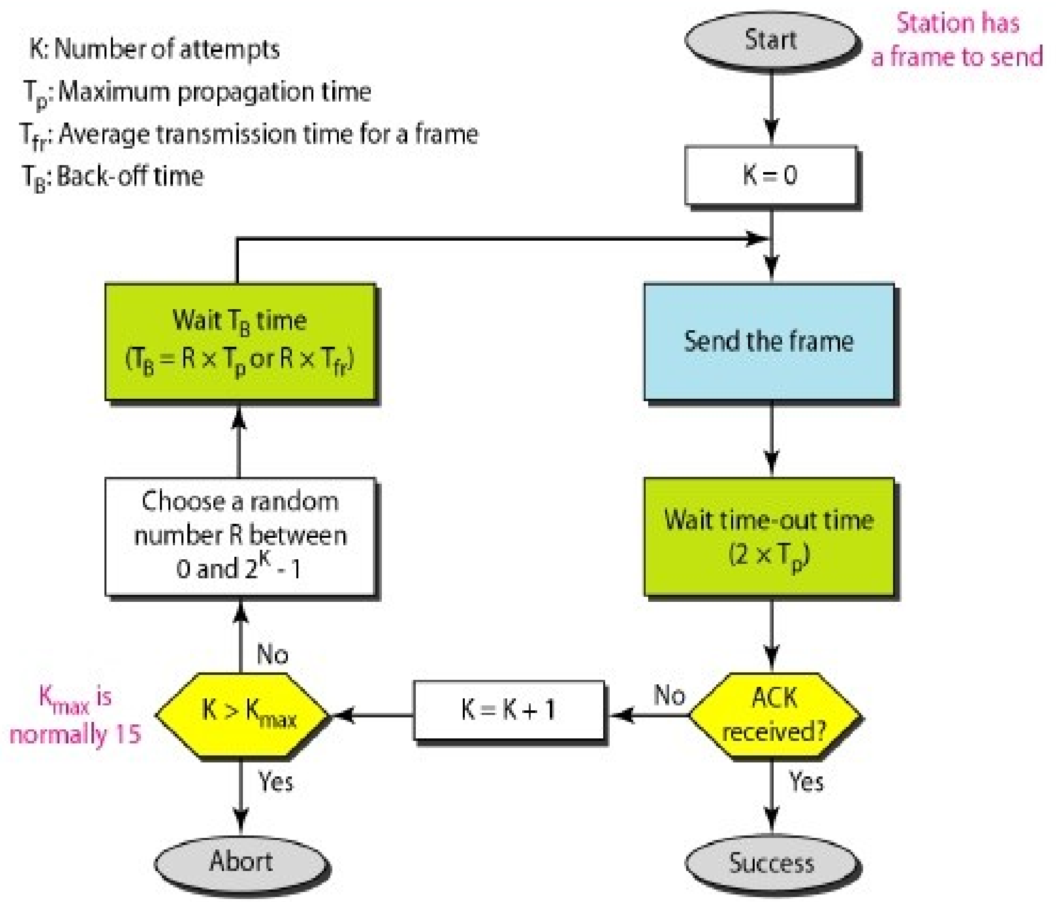
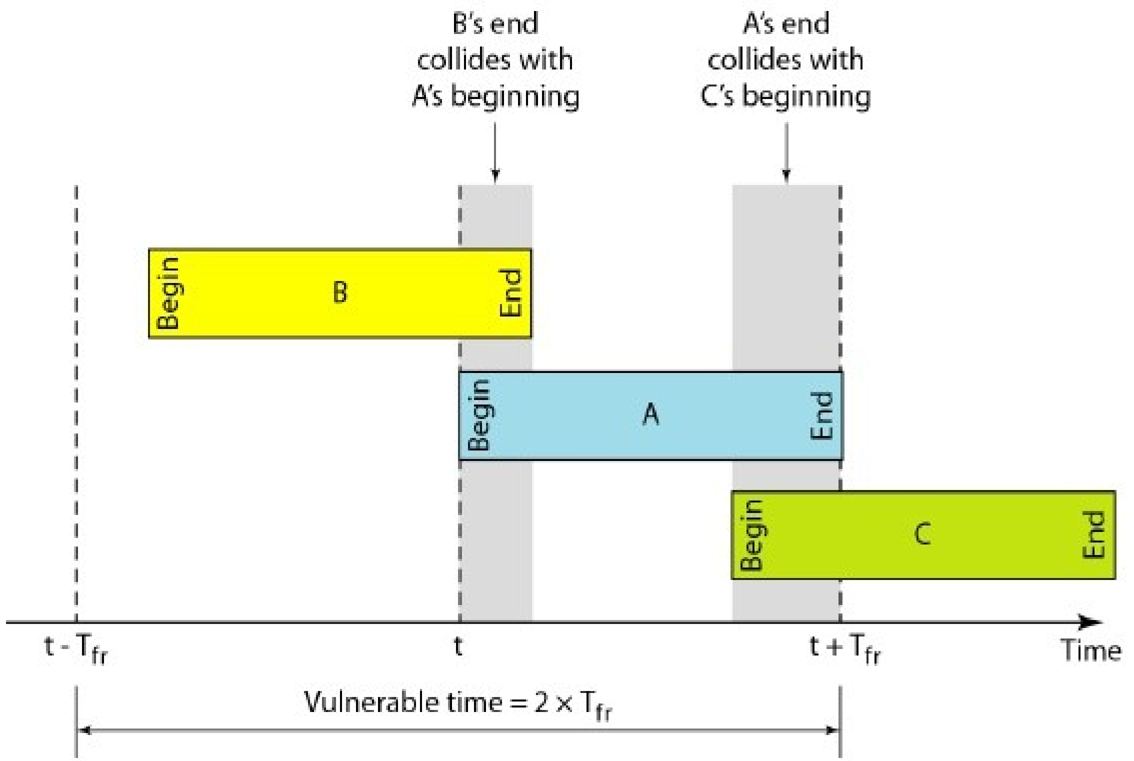


Figure 12.5 ***Vulnerable time for pure ALOHA protocol***



Pure ALOHA

- *Whenever two frames try to occupy the channel at the same time, there will be a collision and both will be garbled.*
- *If the first bit of the new frame overlaps with the last bit of a frame almost finished , both frames will be totally destroyed, and both will be retransmitted later.*
- *Throughput for pure ALOHA decreases.*

Pure ALOHA

Disadvantage

- *More number of users share common channels in a way that can lead to conflicts.*
- *More number of collisions occur.*
- *Collision detected: stations waits a random amount of time.*

Issues with pure aloha

- *Contention, collision detection, retransmission*
- *Vulnerable period for a frame*
- *If G is the mean number of transmission attempts (old and new) per* ***frame time (offered load)****, S is the throughput per frame time, then* $S = Ge^{-2G}$ *for pure ALOHA.*
- *Maximum throughput occurs at G = 0.5 with S=1/(2e)=0.1839. I.e., maximum channel utilization is 18 percent.*

- *In other words, if one-half a frame is generated during one frame transmission time, then 18.4 percent of these frames reach their destination successfully.*
- *This is an expected results because the vulnerable time is 2 times the frame transmission time.*
- *Therefore, if a station generates only one frame in this vulnerable time, the frame will reach its destination successfully.*

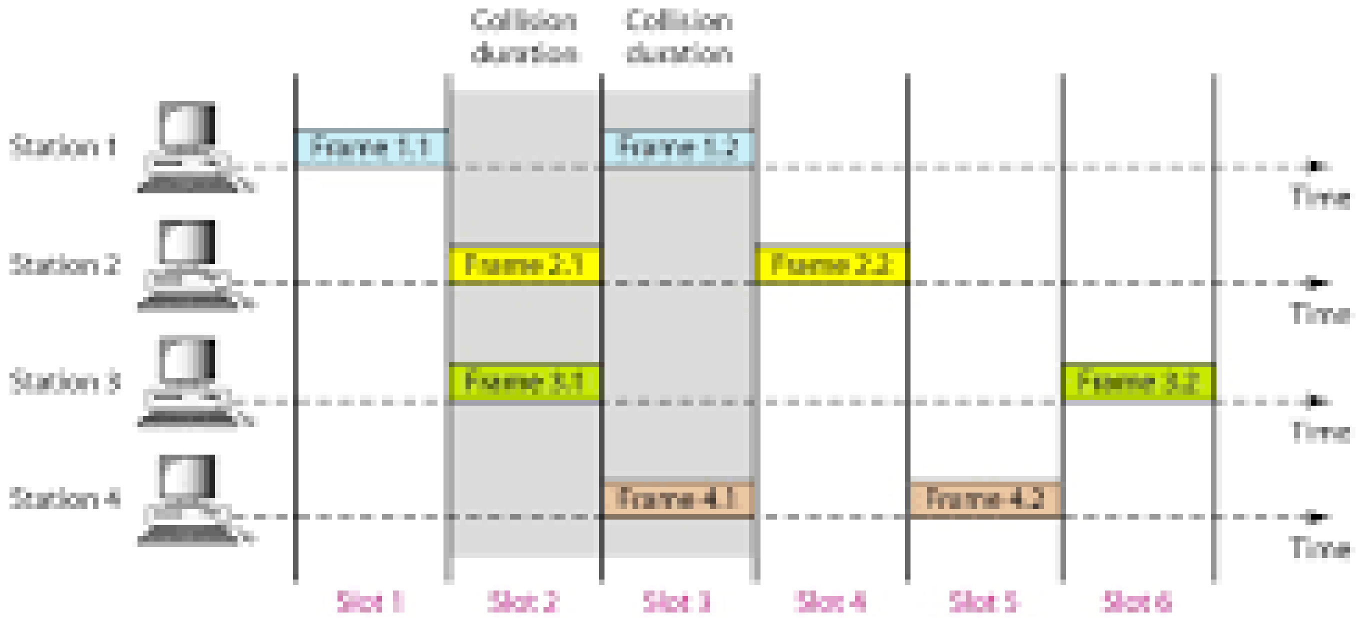
Note

The throughput for pure ALOHA is
$S = G \times e^{-2G}$.
The maximum throughput
$S_{max} = 0.184$ **when G= (1/2).**

Slotted aloha

- *Slotted ALOHA: Divide the time into discrete intervals(slots) of Tfr s, and force the station to send only at the beginning of the time slot..*
- *Obviously, there may be a special signal needed to synchronize the clocks at all stations.*
- *Because a station is allowed to send only at the beginning of synchronized time slot, if a station misses this, it must wait until the beginning of next time slot.*
- *This means that the station which started at the beginning of this slot has already finished sending its frame.*
- *Ofcourse, there is still the possibility of collision if two stations try to send at the beginning of the same slot.*

Figure 12.6 *Frames in a slotted ALOHA network*



- *However, the vulnerable time is now reduced to one-half, equal to T_{fr}.*
- *It can be proved that the average number of successful transmission for the slotted ALOHA is $S=G * e^{-G}$*
- *Maximum throughput occurs at G=1, S=1/e or 0.368. This is twice that of pure ALOHA protocol.*
- *In other words, if a frame is generated during one frame transmission time, then 36.8 % of these frames reach their destination successfully. This result is expected because vulnerable time is equal to the frame transmission time.*
- *Therefore, if a station generates only one frame in this vulnerable time(and no other station generates a frame during this time), the frame will be reach its destination successfully.*

Differences Between Pure ALOHA and Slotted ALOHA

- ***Transmission***
  - *In Pure ALOHA when a frame first arrives, the node immediately transmits the frame in its entirely into the Broadcast Channel.*
  - *In Slotted ALOHA when a node has a fresh frame to send, it waits until the beginning of the next slot and transmits the entire frame in the slot.*
- ***Timing***
  - *In Pure ALOHA Nodes can transmit frames at Random Times.*
  - *In Slotted ALOHA Nodes can transmit frames in their respective slot boundaries only at the beginning of the Slot.*

Differences Between Pure ALOHA and Slotted ALOHA

- ***Synchronization***
  - *Pure ALOHA does not require Synchronization of slots of any nodes.*
  - *Slotted ALOHA requires synchronization between slots of nodes.*
- ***Mode of Transfer***
  - *In Pure ALOHA the Mode of Transfer is Continuous.*
  - *In Slotted ALOHA the mode of transfer is Discrete*

Differences Between Pure ALOHA and Slotted ALOHA

- ***Collission***
  - *In Pure ALOHA If a Collision Occurs the nodes will then immediately retransmit the frame with probability P or the frame transmission time for retransmitting the frame.*
  - *In Slotted ALOHA, if a collision occurs, the node detects the collision before the end of the slot the node retransmits its frame in each subsequent slot with probability P until the frame transmitted without a collision.*
- ***Efficiency***
  - *In Pure ALOHA Efficiency is Half of Slotted ALOHA.*
  - *In Slotted ALOHA efficiency is more than that of Pure ALOHA.*

CSMA: Carrier Sense Multiple Access

- *Protocols in which stations listen for a carrier (i.e., transmission) and act accordingly are called carrier sense protocols.*

Adv:

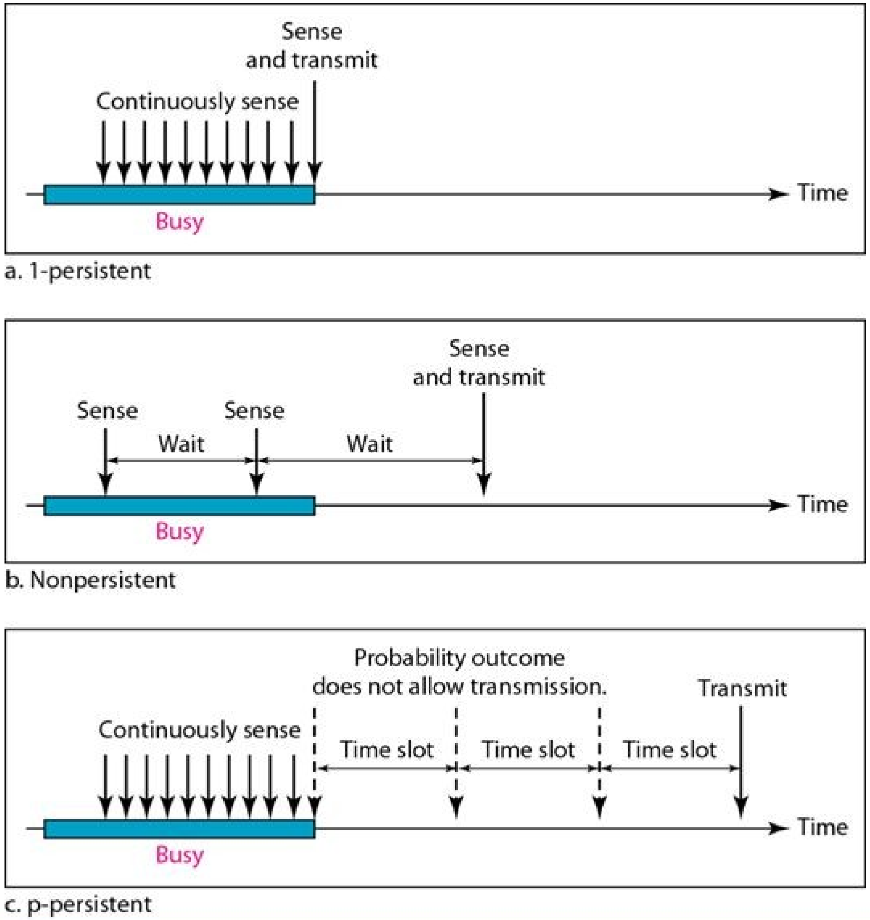
- *To minimize the chance of collision*
- *Increases the performance.*

  - *CSMA principle is "sense before transmit" or "listen before talk".*

CSMA Methods

- *1-persistent CSMA- constant length packets.*
- *non-persistent CSMA- to sense the channel.*
- *p-persistent CSMA*

Figure 12.10 *Behavior of three persistence methods*



1-Persistent-after station finds the line idle, send its frame

Nonpersistent-senses the line; idle: sends immediately; not idle: waits random amount of time and senses again

p-Persistent-the channel has time slots with duration equal to or greater than max propagation time

1-persistent

- *When a station has data to send, it first listens to channel to see if any one else is transmitting at that moment.*
- *If the channel is busy, the station continuously senses the channel until it becomes idle.*
- *When the station detects an idle channel, it transmits a frame.*
- *If a collision occurs, the station waits a random amount of time and starts all over again.*
- *The station transmits with a probability of 1 whenever if finds the channel idle.*
- *This method has highest chance of collision because two or more stations may find the line idle and send their frames immediately.*

Non-persistent CSMA

- *A station that has a frame to send it senses the line.*
- *If the line is idle, it sends immediately.*
- *If the line is not idle, it waits a random amount of time and then senses the line again.*
- *This approach reduces the chance of collision because it is unlikely that two or more stations will wait the same amount of time and retry to send simultaneously.*
- *This algorithm should lead to better channel utilization and longer delays than 1-persistant CSMA.*

P-persistent CSMA

- *This method is used if the channel has time slots with a slot duration equal to or greater than the maximum propogation time.*
- *It combines advantages of the other two strategies.*
- *It reduces the chance of collision and improves efficiency.*

P-persistent CSMA

- *In this method, after station finds the line idle it follows these steps:*

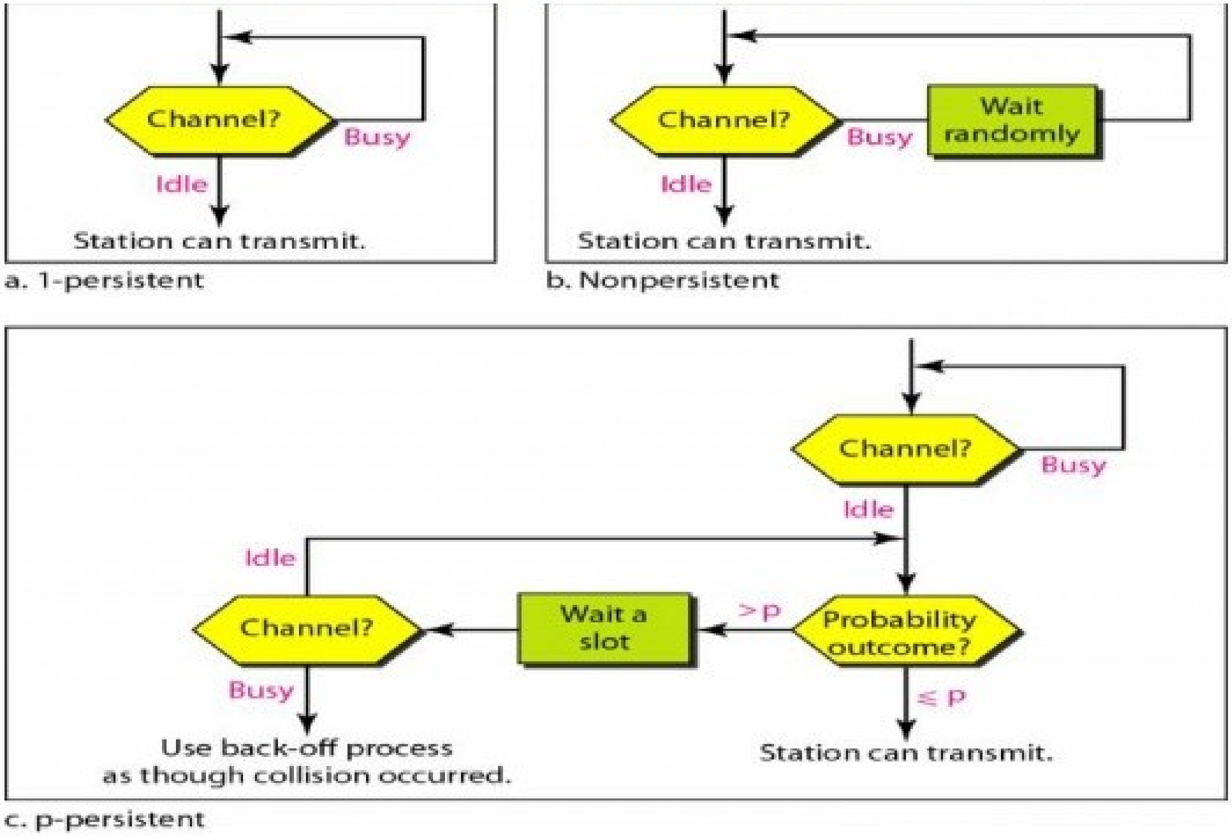
1.With probability 'p', the station sends its frame.

2.With probability q=1-p, the station waits for the beginning of the next time slot and checks the line again.

a. If the line is idle, it goes to step 1.

b. If the line is busy, it acts as though a collision has occurred and uses the back-off procedure(which discussed earlier).

Figure 2.39 shows the flow diagrams for these methods.



Carrier Sense Multiple Access with Collision Detection(CSMA/CD)

- In this method, a station monitors the medium after it sends a frame to see if the transmission was successful. If so, the station is finished. If, however, there is a collision, the frame is sent again.

CSMA/CD

- (CSMA/CD) augments the algorithm to handle the collision.
- If, however, there is a collision, the frame is sent again.

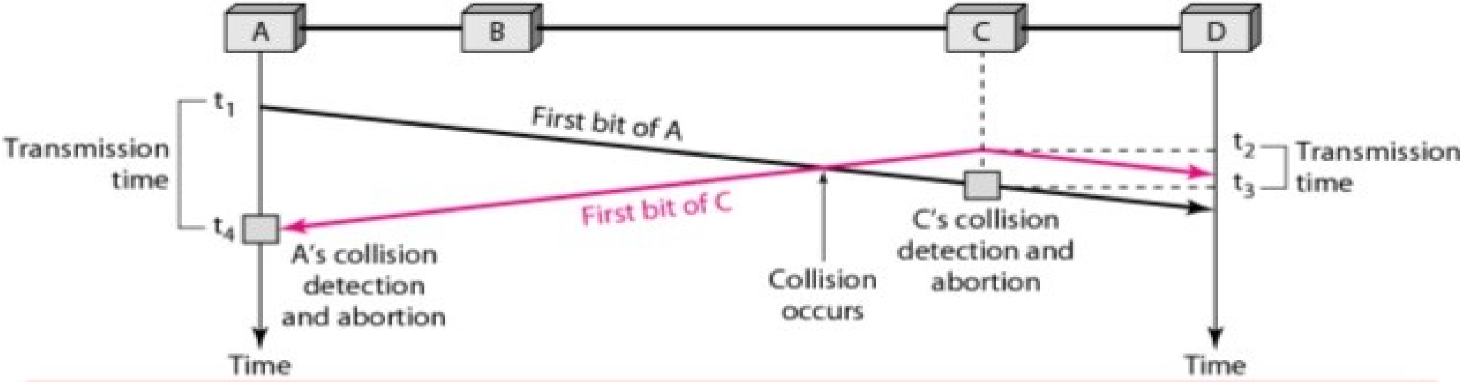
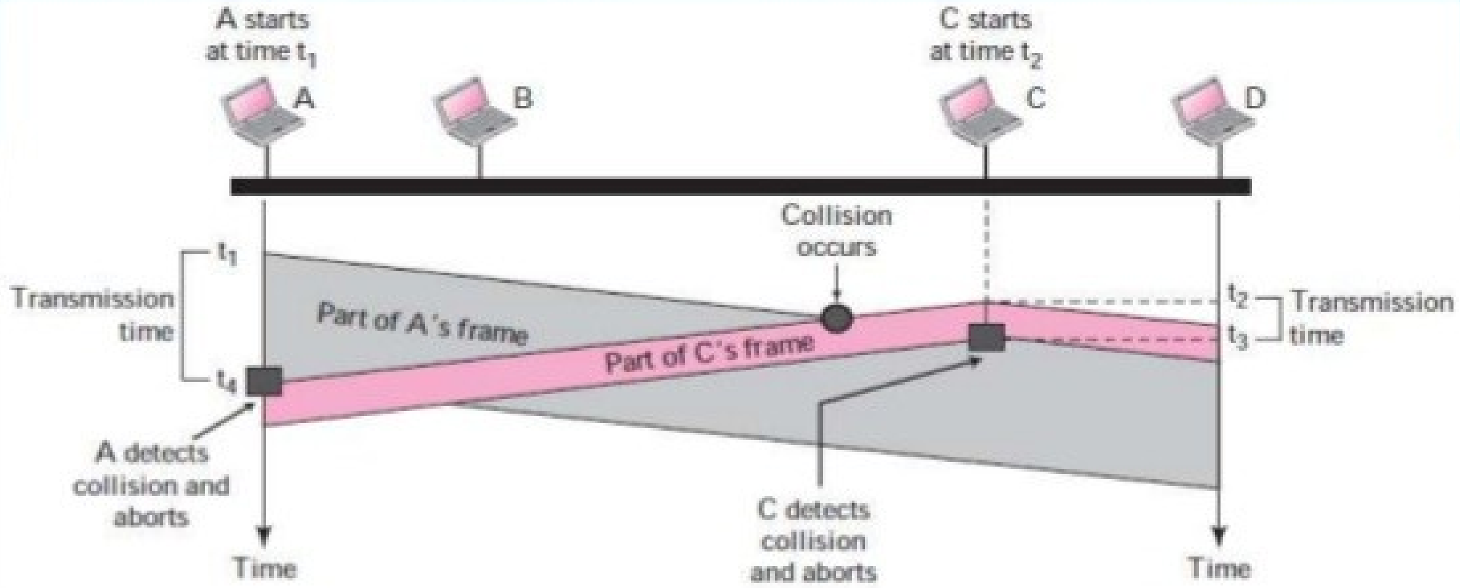


 Figure 12.12 *Collision of the first bit in CSMA/CD*



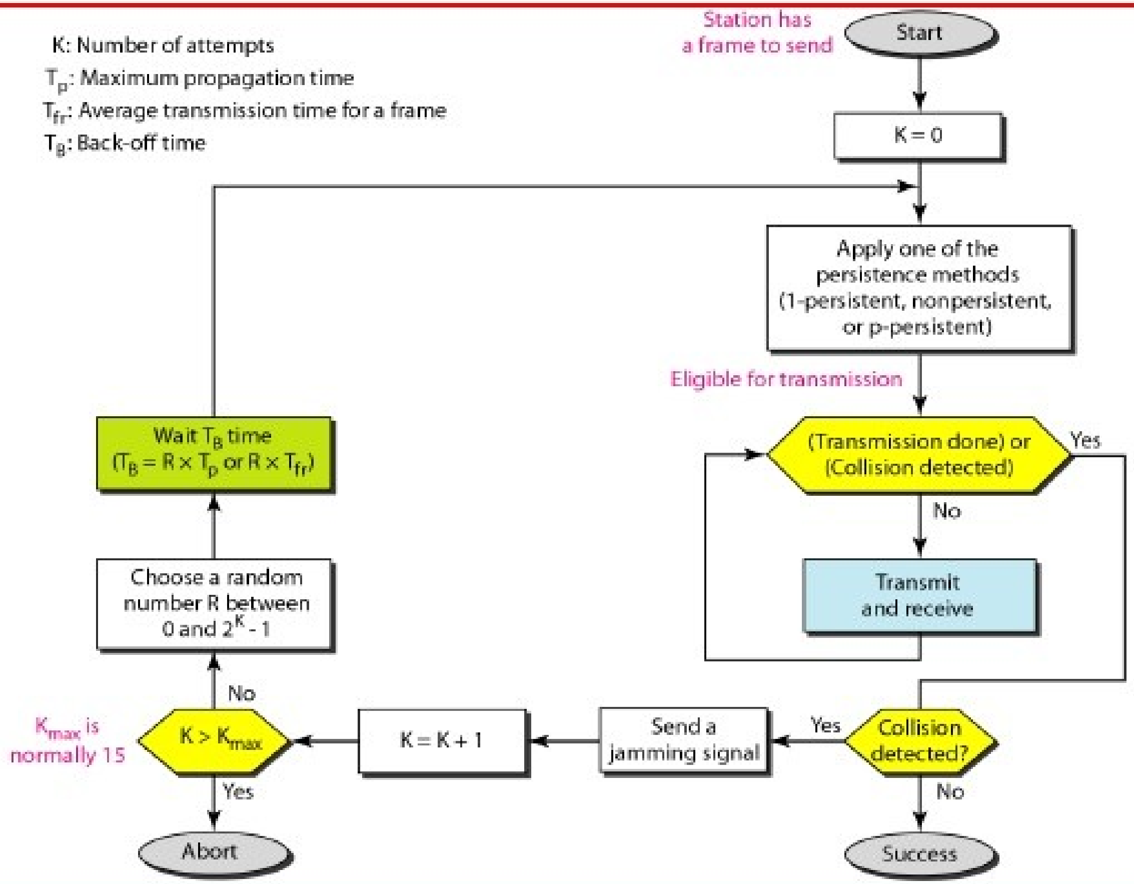
Collision of the first bit in CSMA/CD

- A transmits for the duration t4 - t1; C transmits for the duration t3 - t2 → for the protocol to work, the length of any frame divided by the bit rate in this protocol must be more than either of these durations
- Before sending the last bit of the frame, the sending station must detect a collision, if any, and abort the transmission → because, once the entire frame is sent, station does not keep a copy of the frame and does not monitor the line for collision detection

Minimum Frame Size

- *For CSMA/CD to work, we need restriction on the frame size.*
- *Therefore, the frame transmission time Tfr must be at least two times the maximum propagation time Tp.*
- *To understand the reason, let us think about worst-case scenario. If two stations involved in a collision are the maximum distance apart, the signal from the first takes Tp to reach the second, and the effect of the collision takes another Tp to reach the first.*
- *So the requirement is that the first station must still be transmitting after 2Tp*

Figure 12.14 *Flow diagram for the CSMA/CD*



CSMA/CD Vs ALOHA

- CSMA/CD is similar to the one for ALOHA protocol, but there are differences which as follows:

1. Addition of persistence methods
2. Frame transmission: In ALOHA, we first transmit the entire frame and then wait for an acknowledgment.

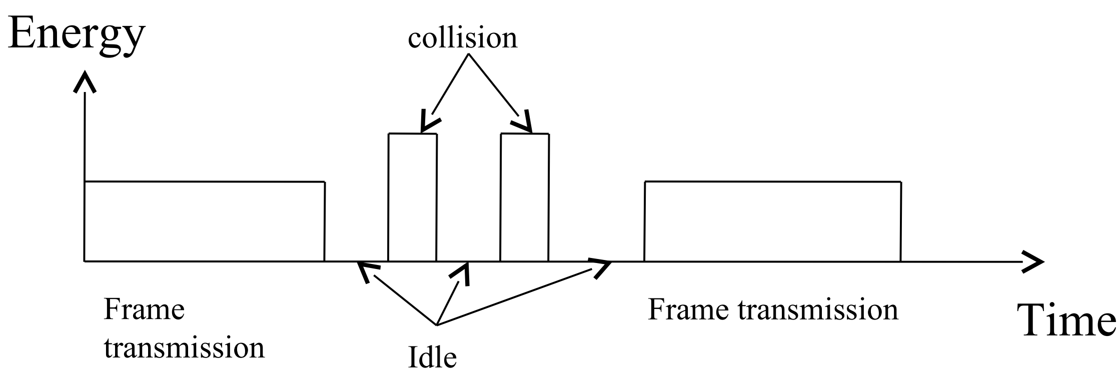
   In CSMA/CD, transmission and collision is a continuous process. We do not send entire frame and then look for a collision. The station transmits and receives continuously and simultaneously.

Contd..

- We use loop to show that transmission is a continuous process.
- We constantly monitor in order to detect one of the two conditions: either transmission is finished or a collision is detected. Either event stops transmission.
- When we come out of loop, if a collision has not detected, it means that the transmission is complete: the entire frame is transmitted. Otherwise, a collision has occurred.

3. Sends a short Jamming signal that enforces the collision in case other stations have not yet sensed the collision

Energy level in CSMA/CD



Throughput of CSMA/CD

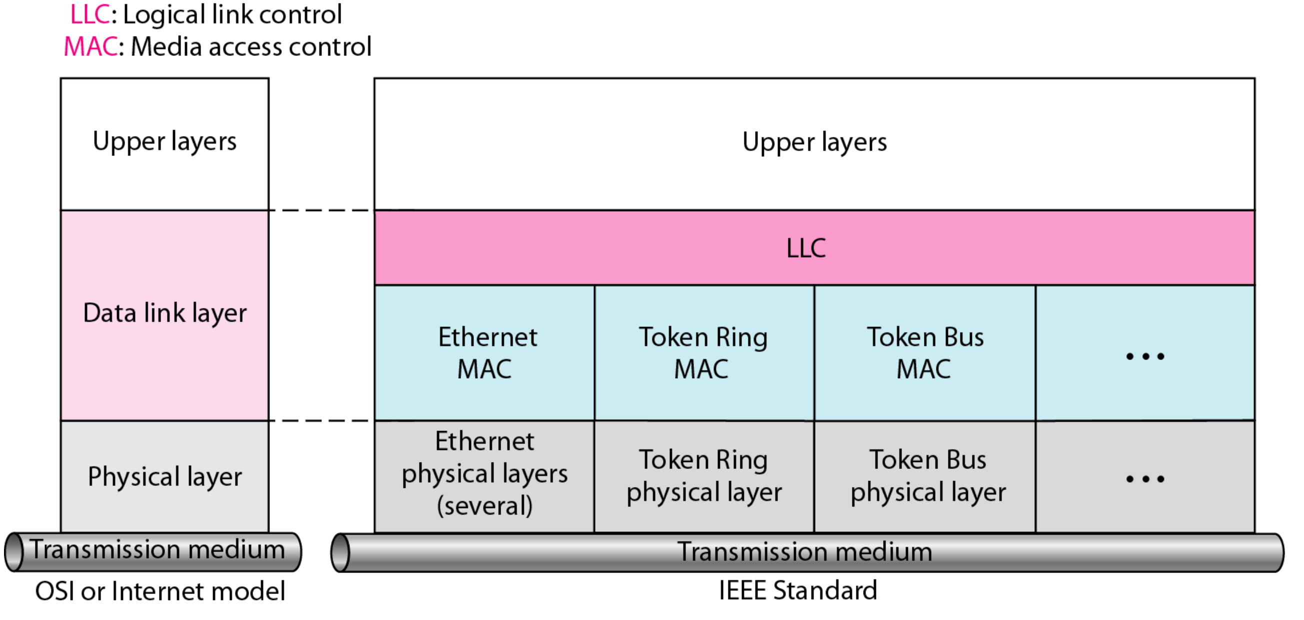
- The throughput of CSMA/CD is greater than that of pure or slotted ALOHA.
- The maximum throughput occurs at a different value of G and is based on the persistent method and the value of p in the p-persistent approach.
- For 1-persistent method the maximum throughput is around 50 % when G=1.
- For non-persistent method, the maximum throughput can go up to 90 % when G is between 3 and 8.

Wired LANs: *Ethernet*

1. IEEE Standards
2. Standard Ethernet
3. Changes in the Standard
4. Fast Ethernet
5. Gigabit Ethernet

IEEE Standards

- In 1985, the Computer Society of the IEEE started a project, called Project 802, to set standards to enable intercommunication among equipment from a variety of manufacturers. Project 802 is a way of specifying functions of the physical layer and the data link layer of major LAN protocols.

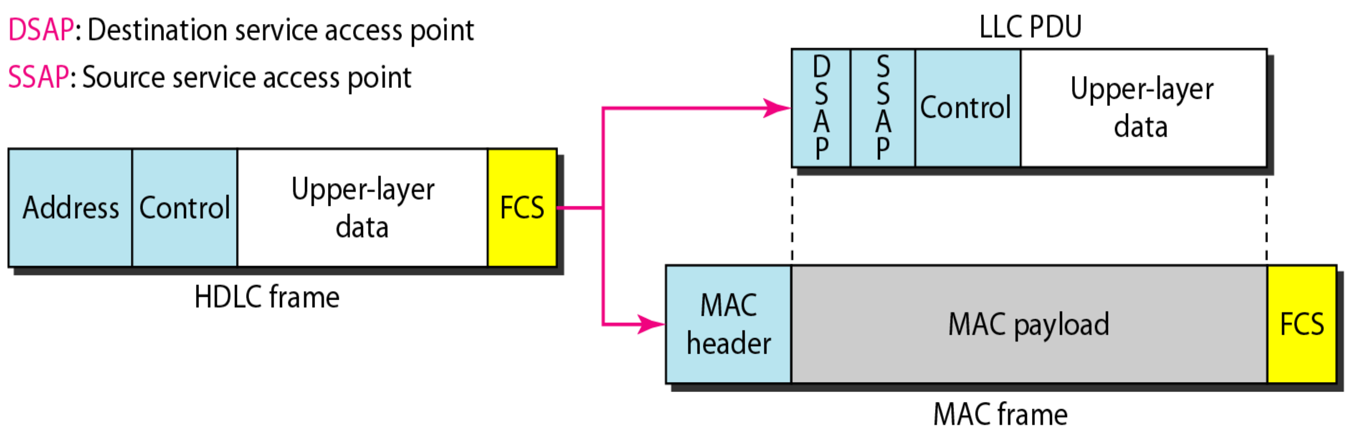


IEEE 802 Working Group

Active working groups	Inactive or disbanded working groups
802.1 Higher Layer LAN Protocols Working Group	802.2 Logical Link Control Working Group
802.3 Ethernet Working Group	802.4 Token Bus Working Group
802.11 Wireless LAN Working Group	802.5 Token Ring Working Group
802.15 Wireless Personal Area Network (WPAN) Working Group	802.7 Broadband Area Network Working Group
802.16 Broadband Wireless Access Working Group	802.8 Fiber Optic TAG
802.17 Resilient Packet Ring Working Group	802.9 Integrated Service LAN Working Group
802.18 Radio Regulatory TAG	802.10 Security Working Group
802.19 Coexistence TAG	802.12 Demand Priority Working Group
802.20 Mobile Broadband Wireless Access (MBWA) Working Group	802.14 Cable Modem Working Group
802.21 Media Independent Handoff Working Group	
802.22 Wireless Regional Area Networks	

Logical Link Control (LLC)

- Framing: LLC defines a protocol data unit (PDU) that is similar to that of HDLC
- To provide flow and error control for the upper-layer protocols that actually demand these services.
- It provides one single data link control protocol for all IEEE LANs.



Medium Access Control(MAC)

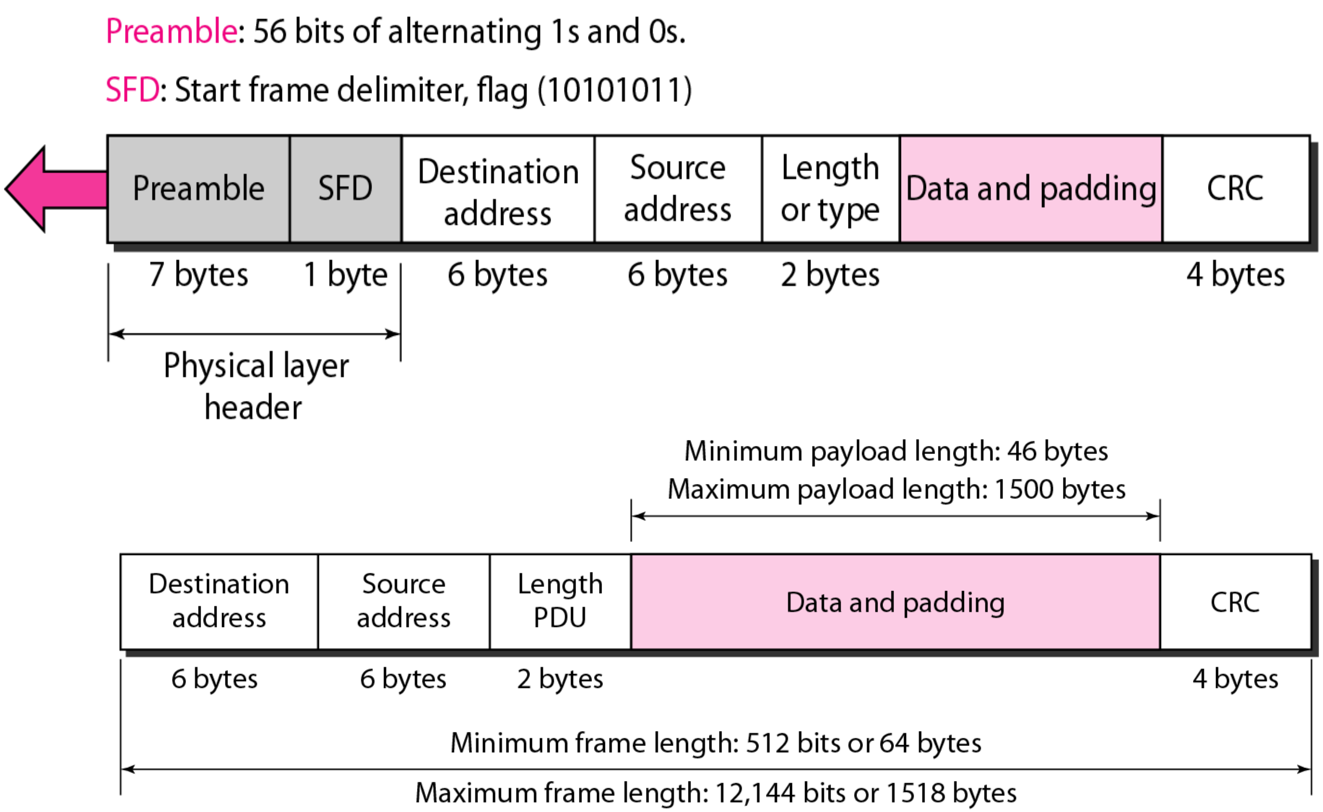
- IEEE Project 802 has created a sub-layer called MAC that defines specific access methods for each LAN.
- In contrast to LLC sub-layer, the MAC sub layer contains a number of distinct modules; each defines the access method and the framing format specific to the LAN protocol.

Physical layer

- The physical layer is dependent on the implementation and the type of physical media used.

MAC Sublayer

- Preamble: Alternate 0's and 1's that alerts the receiving system to the coming frame and enables it to synchronize its input timing

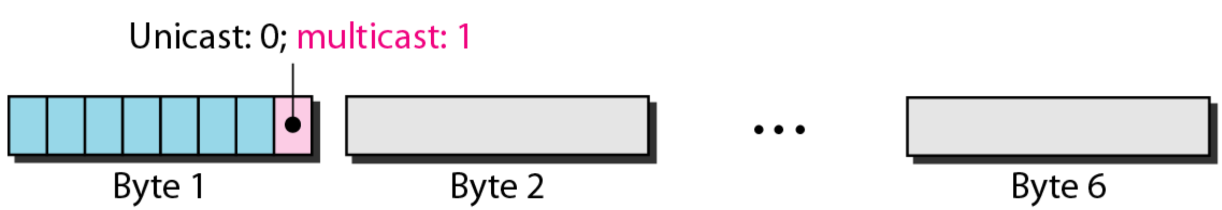


Addressing

- Ethernet address in hexadecimal notation

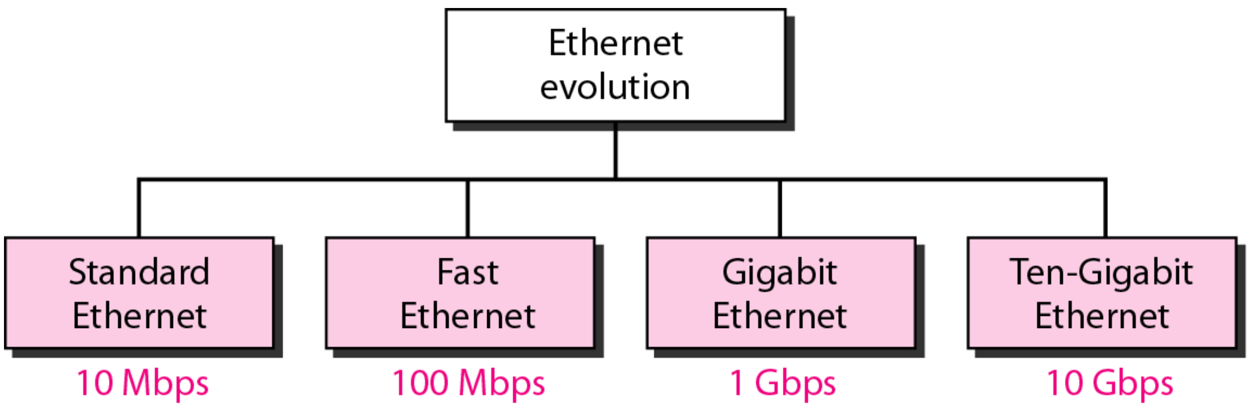
$$\underbrace{06 : 01 : 02 : 01 : 2C : 4B}_{\text{6 bytes = 12 hex digits = 48 bits}}$$

- The least significant bit of the first byte defines the type of address. If the bit is 0, the address is unicast; otherwise, it is multicast
- The broadcast destination address is a special case of the multicast address in which all bits are 1s

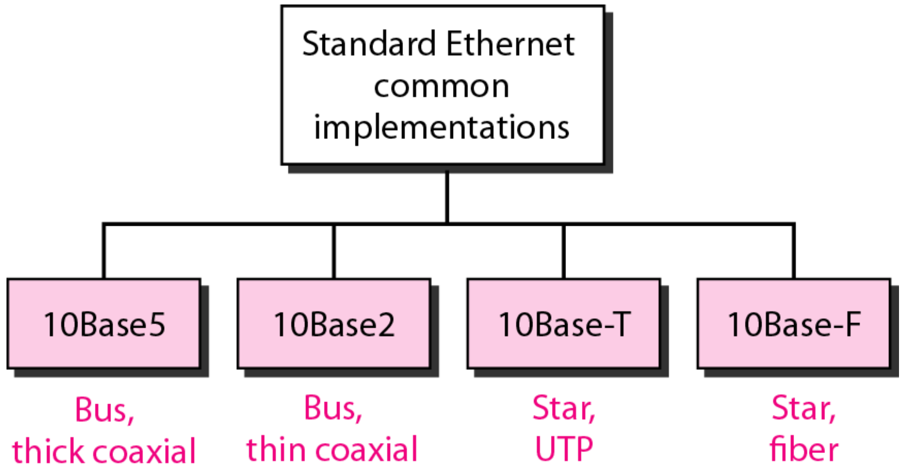


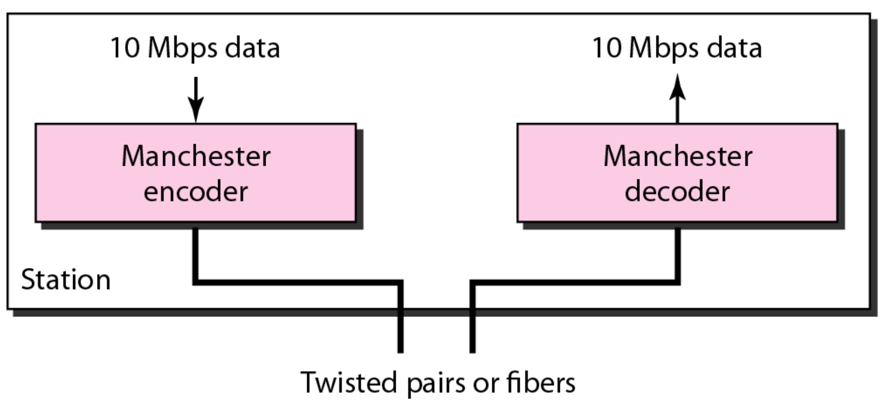
Standard Ethernet

- The original Ethernet was created in 1976 at Xerox's Palo Alto Research Center (PARC). Since then, it has gone through four generations

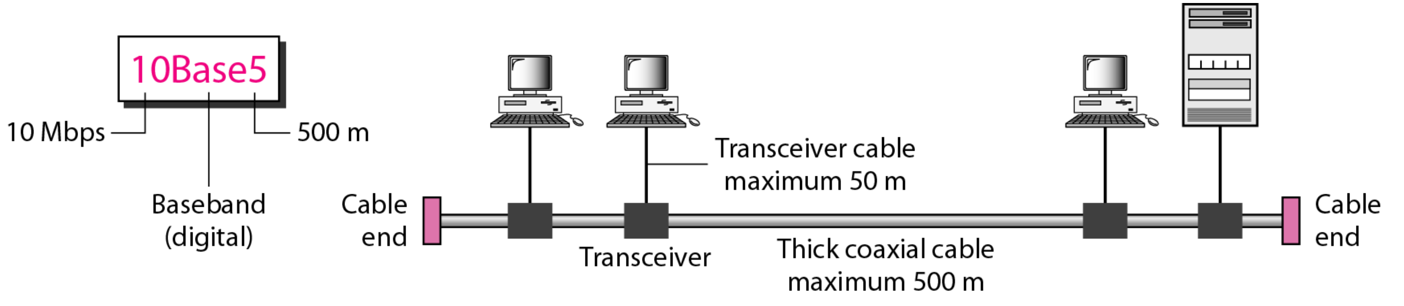


Physical Layer: Ethernet

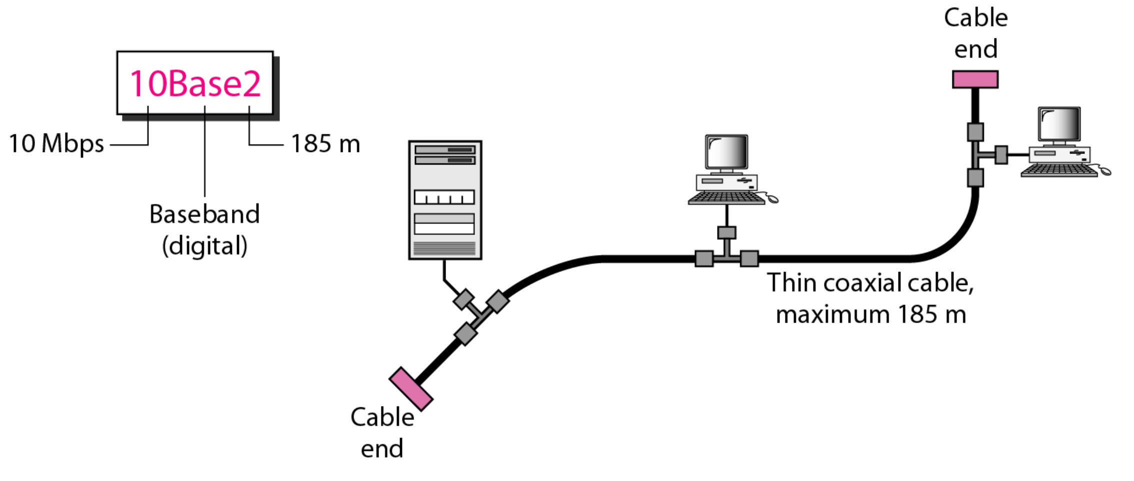




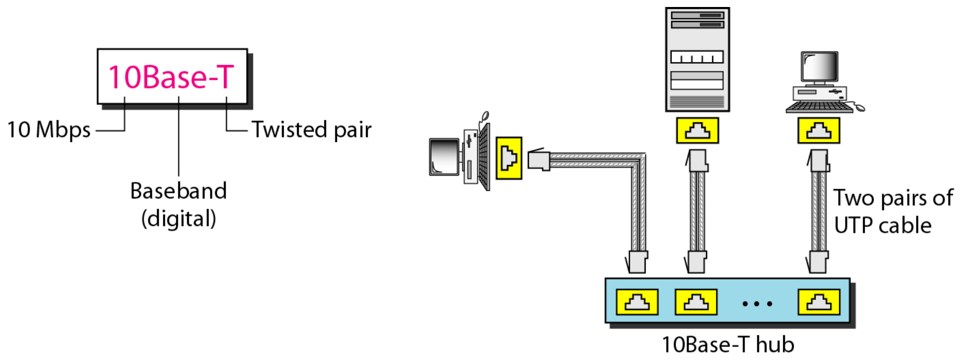
10Base5: Thick Ethernet



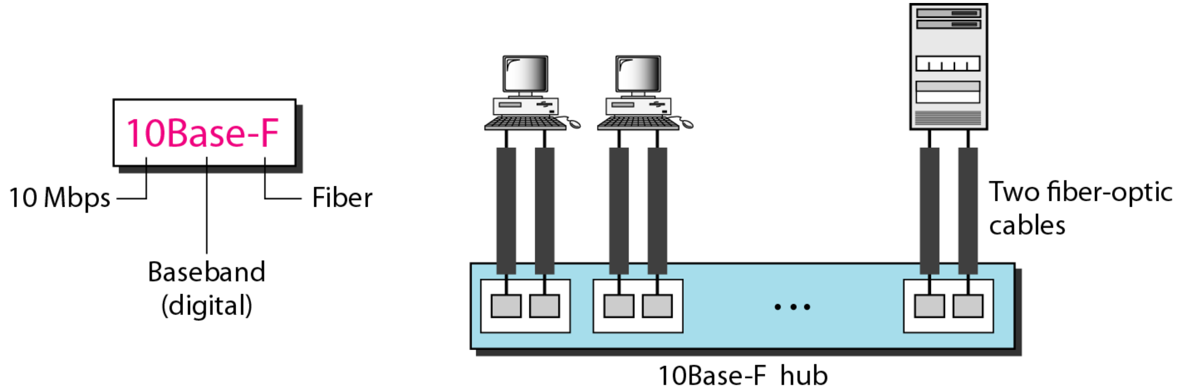
10Base2: Thin Ethernet



10BaseT: Twisted-Pair Ethernet



10Base-F: Fiber Ethernet

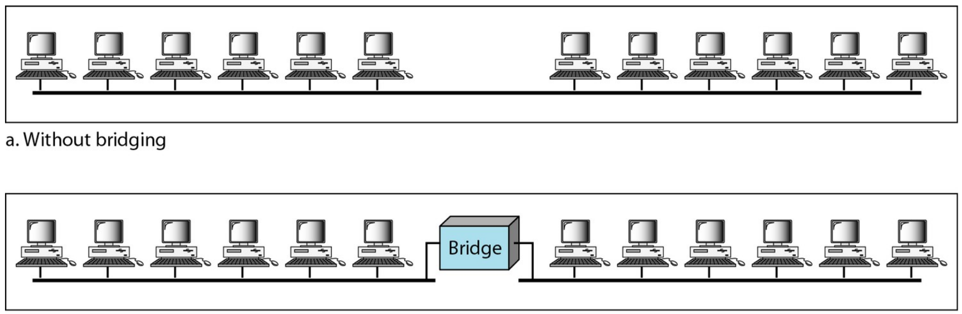


Summary of Standard Ethernet

Characteristics	*10Base5*	*10Base2*	*10Base-T*	*10Base-F*
Media	Thick coaxial cable	Thin coaxial cable	2 UTP	2 Fiber
Maximum length	500 m	185 m	100 m	2000 m
Line encoding	Manchester	Manchester	Manchester	Manchester

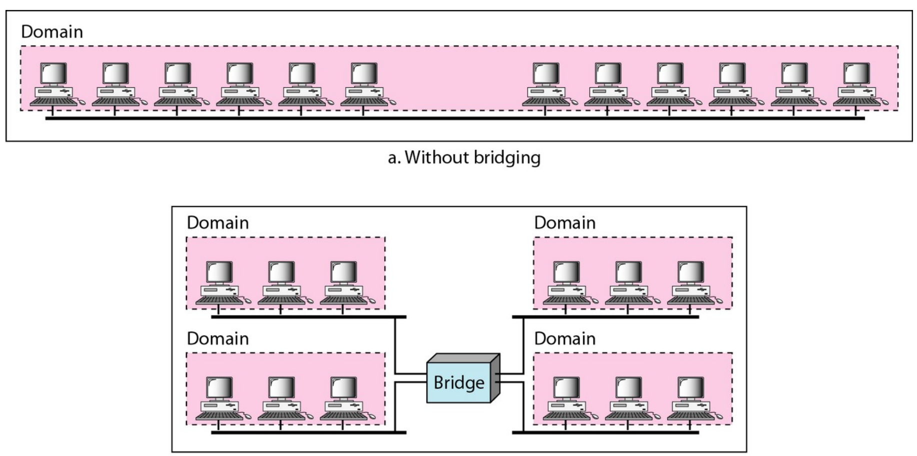
Changes in the Standard

- Bridged Ethernet: Raising bandwidth and separating collision domains



a. Without bridging

b. With bridging

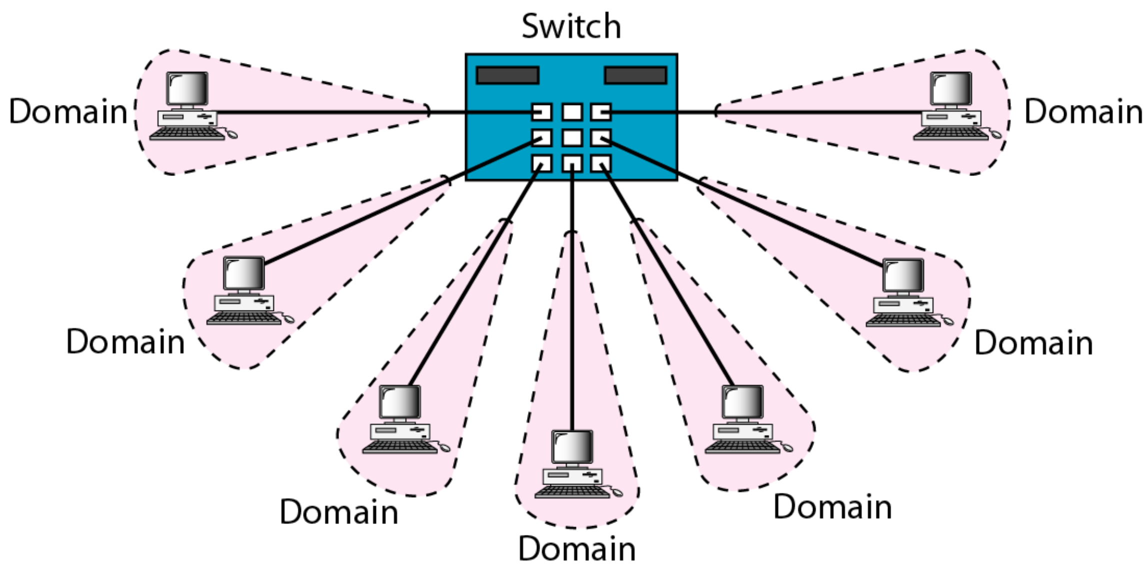


a. Without bridging

b. With bridging

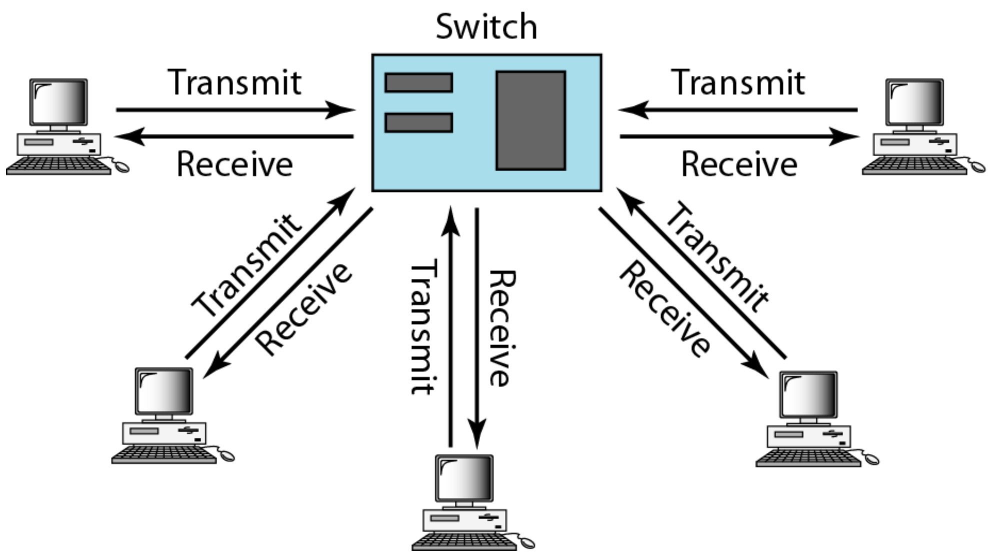
Changes in the Standard

- Switched Ethernet: N-port bridge



Changes in the Standard

- Full-duplex (switched) Ethernet: no need for CSMA/CD



Fast Ethernet

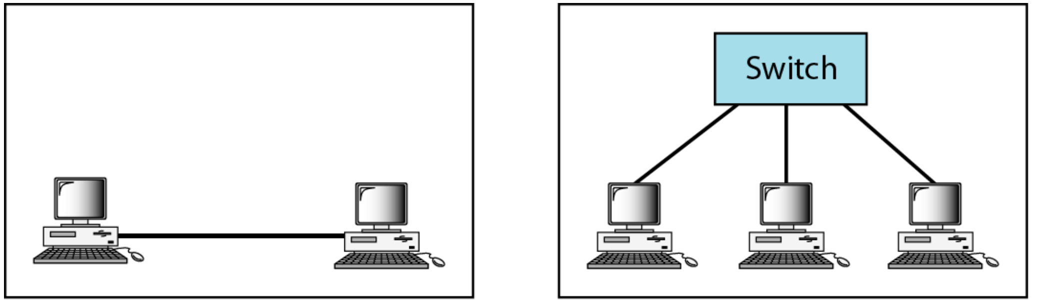
- Under the name of IEEE 802.3u
  - Upgrade the data rate to 100 Mbps
  - Make it compatible with Standard Ethernet
  - Keep the same 48-bit address and the same frame format
  - Keep the same min. and max. frame length

- MAC Sublayer
  - CSMA/CD for the half-duplex approach
  - No need for CSMA/CD for full-duplex Fast Ethernet

- Autonegotiation: allow two devices to negotiate the mode or data rate of operation

Fast Ethernet: Physical Layer

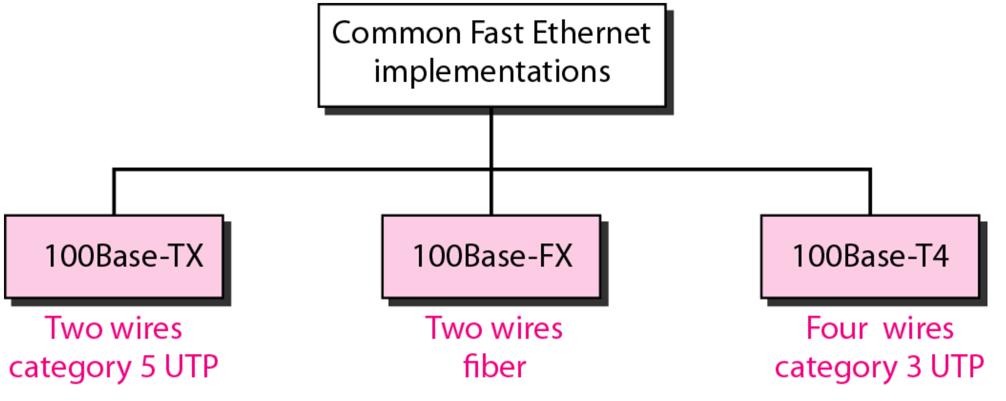
- Topology



a. Point-to-point

b. Star

- Implementation



Summary of Fast Ethernet

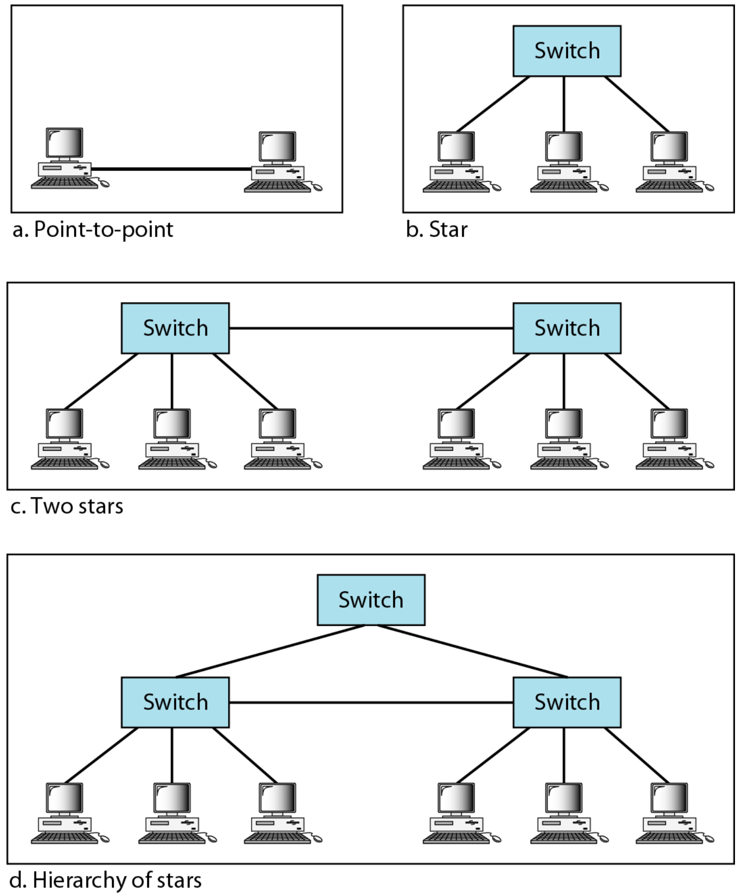
Characteristics	*100Base-TX*	*100Base-FX*	*100Base-T4*
Media	Cat 5 UTP or STP	Fiber	Cat 4 UTP
Number of wires	2	2	4
Maximum length	100 m	100 m	100 m
Block encoding	4B/5B	4B/5B	
Line encoding	MLT-3	NRZ-I	8B/6T

Gigabit Ethernet

- Under the name of IEEE 802.3z
  - Upgrade the data rate to 1 Gbps
  - Make it compatible with Standard or Fast Ethernet
  - Keep the same 48-bit address and the same frame format
  - Keep the same min. and max. frame length
  - Support autonegotiation as defined in Fast Ethernet
- MAC Sublayer
  - Most of all implmentations follows full-duplex approach
  - In the full-duplex mode of Gigabit Ethernet, there is no collision; the maximum length of the cable is determined by the signal attenuation in the cable.
- Half-duplex mode (very rare)
  - Traditional: 0.512 μs (25m)
  - Carrier Extension: 512 bytes (4096 bits) min. length
  - Frame bursting to improve the inefficiency of carrier extension

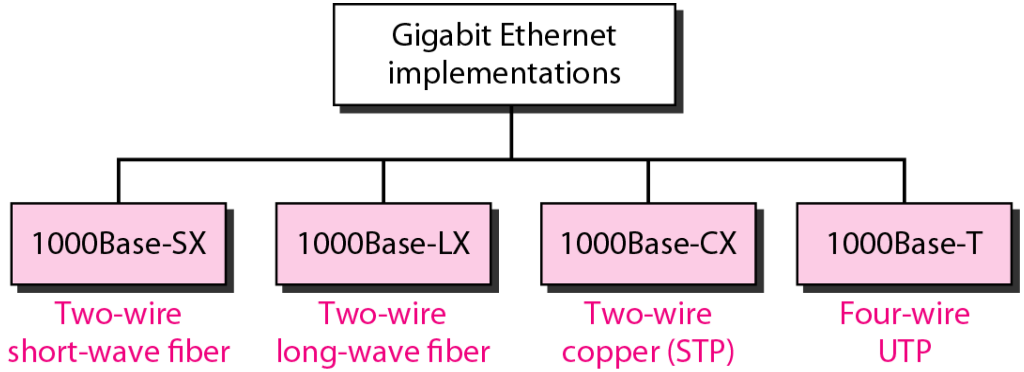
Gigabit Ethernet: Physical Layer

- Topology



Gigabit Ethernet: Physical Layer

- Implementation



Gigabit Ethernet: Summary

Characteristics	*1000Base-SX*	*1000Base-LX*	*1000Base-CX*	*1000Base-T*
Media	Fiber short-wave	Fiber long-wave	STP	Cat 5 UTP
Number of wires	2	2	2	4
Maximum length	550 m	5000 m	25 m	100 m
Block encoding	8B/10B	8B/10B	8B/10B	
Line encoding	NRZ	NRZ	NRZ	4D-PAM5

Ten-Gigabit Ethernet

- Under the name of IEEE 802.3ae
  - Upgrade the data rate to 10 Gbps
  - Make it compatible with Standard, Fast, and Giga Ethernet
  - Keep the same 48-bit address and the same frame format
  - Keep the same min. and max. frame length
  - Allow the interconnection of existing LANs into a MAN or WAN
  - Make Ethernet compatible with Frame Relay and ATM
- MAC Sublayer: Only in full-duplex mode → no CSMA/CD

Characteristics	*10GBase-S*	*10GBase-L*	*10GBase-E*
Media	Short-wave 850-nm multimode	Long-wave 1310-nm single mode	Extended 1550-mm single mode
Maximum length	300 m	10 km	40 km

Wireless
Local Area Networks

Wireless Local Area Networks

- The proliferation of laptop computers and other mobile devices (PDAs and cell phones) created an *obvious* application level demand for wireless local area networking.
- Companies jumped in, quickly developing *incompatible* wireless products in the 1990's.
- Industry decided to entrust standardization to IEEE committee that dealt with wired LANS – ***namely, the IEEE 802 committee!!***

IEEE 802 Standards Working Groups

Number	Topic
802.1	Overview and architecture of LANs
802.2 ↓	Logical link control
802.3 *	Ethernet
802.4 ↓	Token bus (was briefly used in manufacturing plants)
802.5	Token ring (IBM's entry into the LAN world)
802.6 ↓	Dual queue dual bus (early metropolitan area network)
802.7 ↓	Technical advisory group on broadband technologies
802.8 †	Technical advisory group on fiber optic technologies
802.9 ↓	Isochronous LANs (for real-time applications)
802.10 ↓	Virtual LANs and security
802.11 *	Wireless LANs
802.12 ↓	Demand priority (Hewlett-Packard's AnyLAN)
802.13	Unlucky number. Nobody wanted it
802.14 ↓	Cable modems (defunct: an industry consortium got there first)
802.15 *	Personal area networks (Bluetooth)
802.16 *	Broadband wireless
802.17	Resilient packet ring

Categories of Wireless Networks

- ***Base Station*** :: all communication through an ***access point*** {note hub topology}. Other nodes can be fixed or mobile.
- ***Infrastructure Wireless*** :: base station network is connected to the wired Internet.
- ***Ad hoc Wireless*** :: wireless nodes communicate directly with one another.

Basic service sets (BSSs)

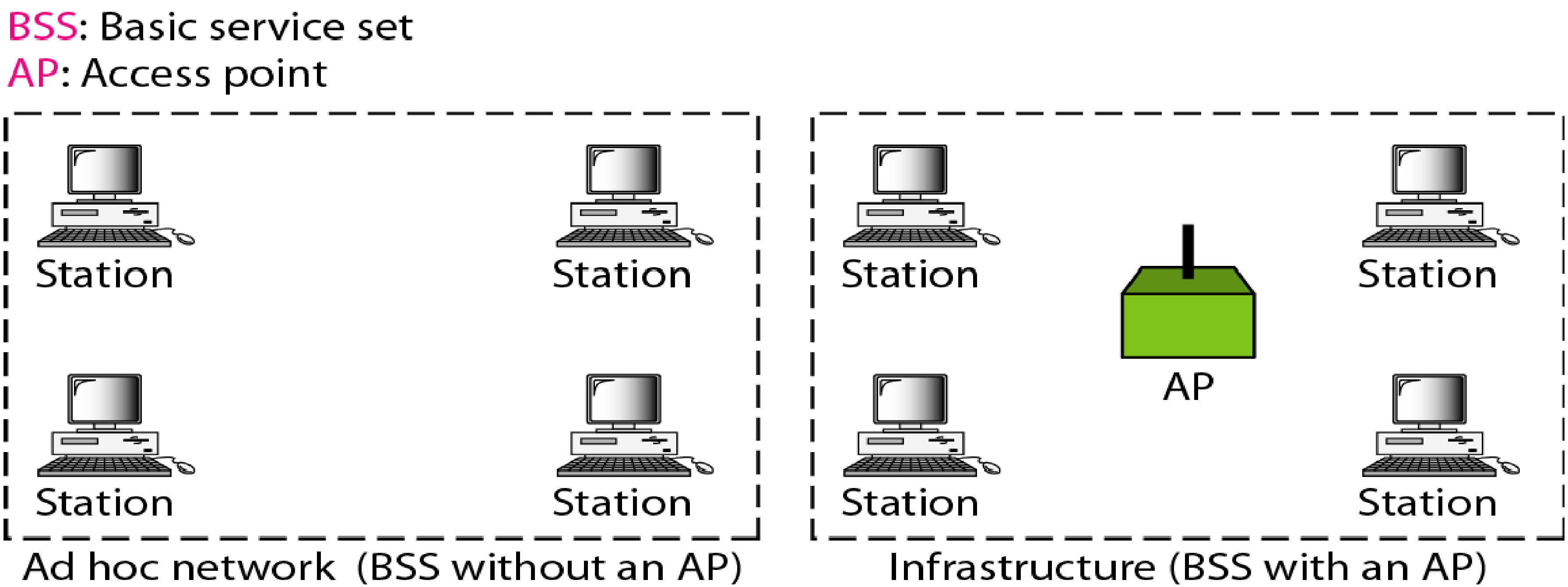
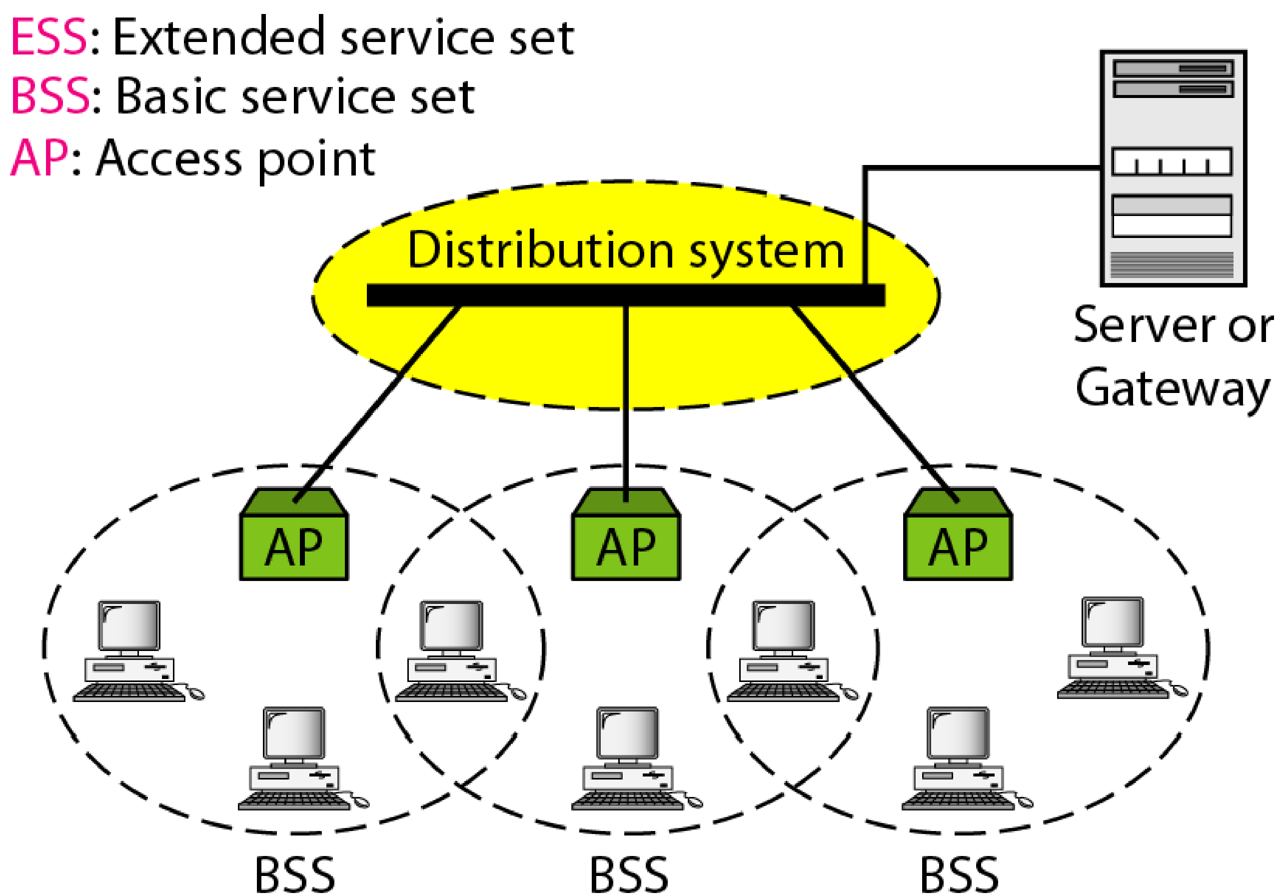
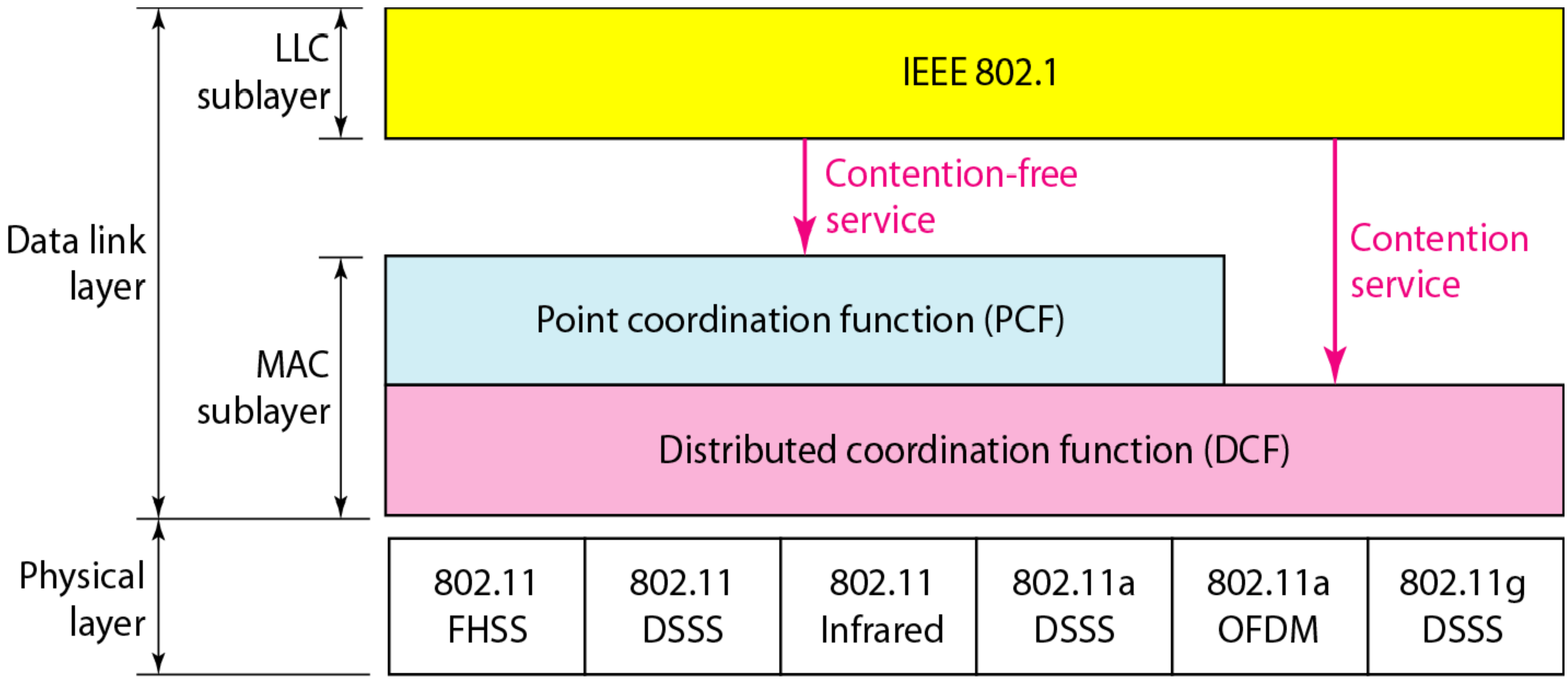


Figure 14.2 Extended service sets (ESSs)



MAC layers in IEEE 802.11 standard



Distribute Coordination Function (DCF)

- Uses CSMA/ CA (CSMA with Collision Avoidance).
  - Uses both physical and ***virtual*** carrier sensing.
  - Two methods are supported:

1. **based on MACAW(Multiple Access with Collision Avoidance for Wireless) with virtual carrier sensing.**
2. **1-persistent physical carrier sensing.**

CSMA/CA

flowchart

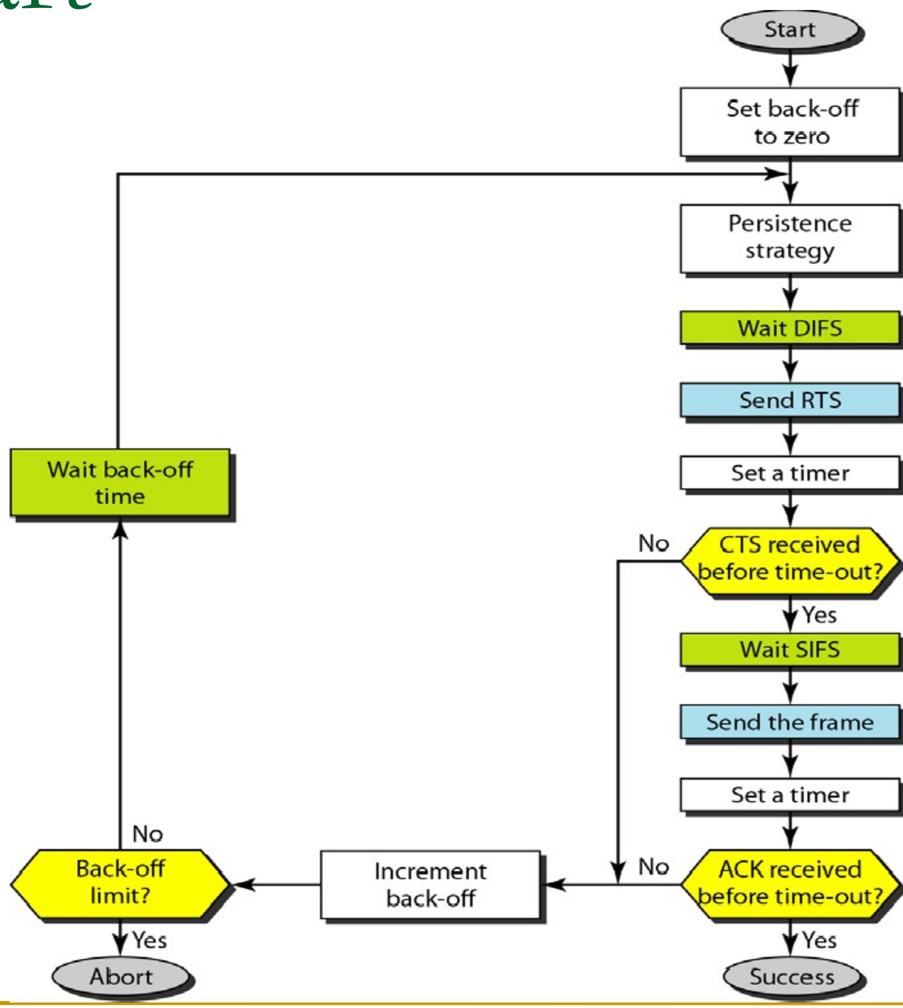
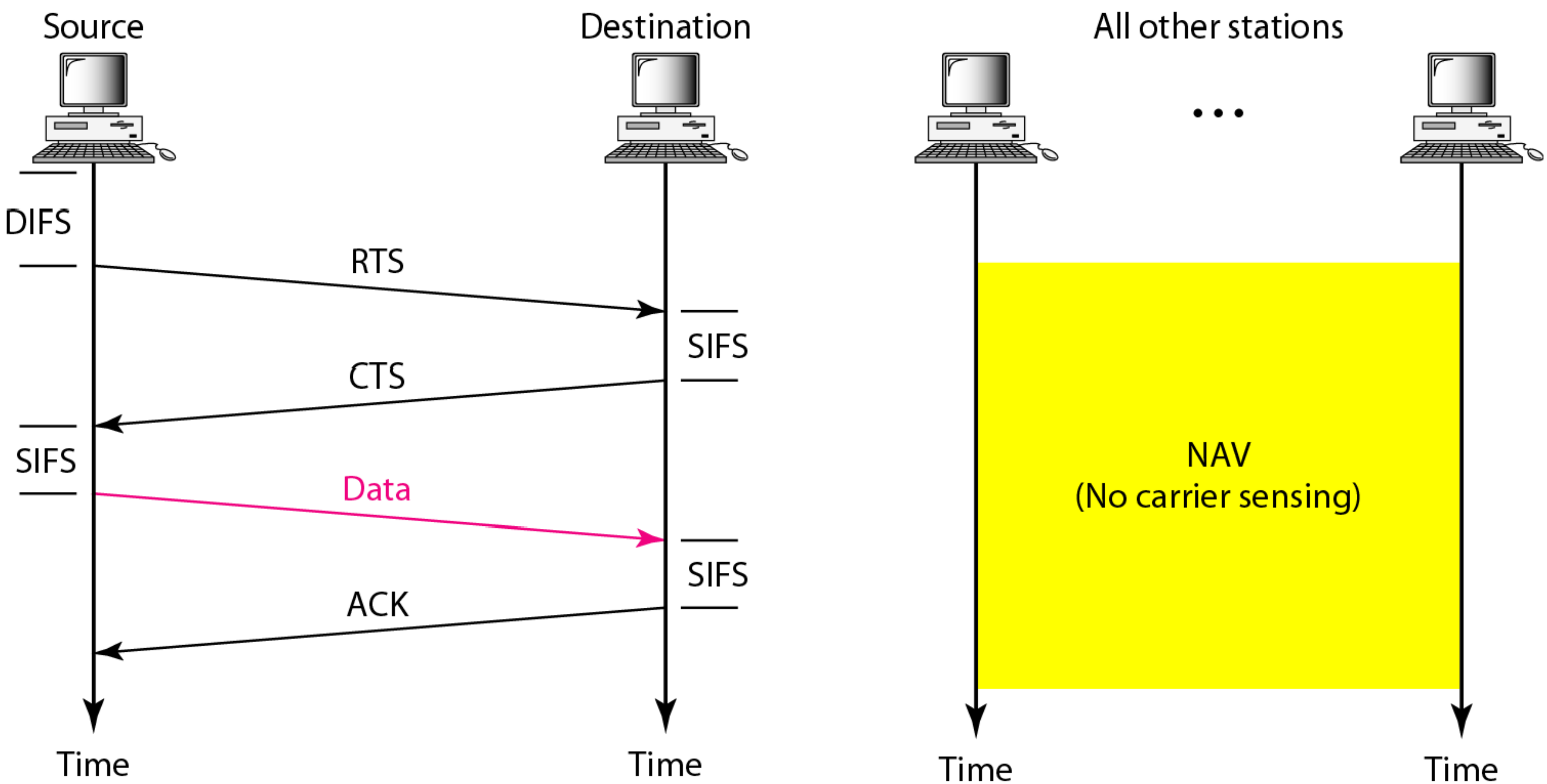


Figure 14.5 CSMA/CA and NAV



Point Coordinated Function (PCF)

- PCF uses a base station to poll other stations to see if they have frames to send.
- No collisions occur.
- Base station sends ***beacon frame*** periodically.
- Base station can tell another station to ***sleep*** to save on batteries and base stations holds frames for sleeping station.

Example of repetition interval

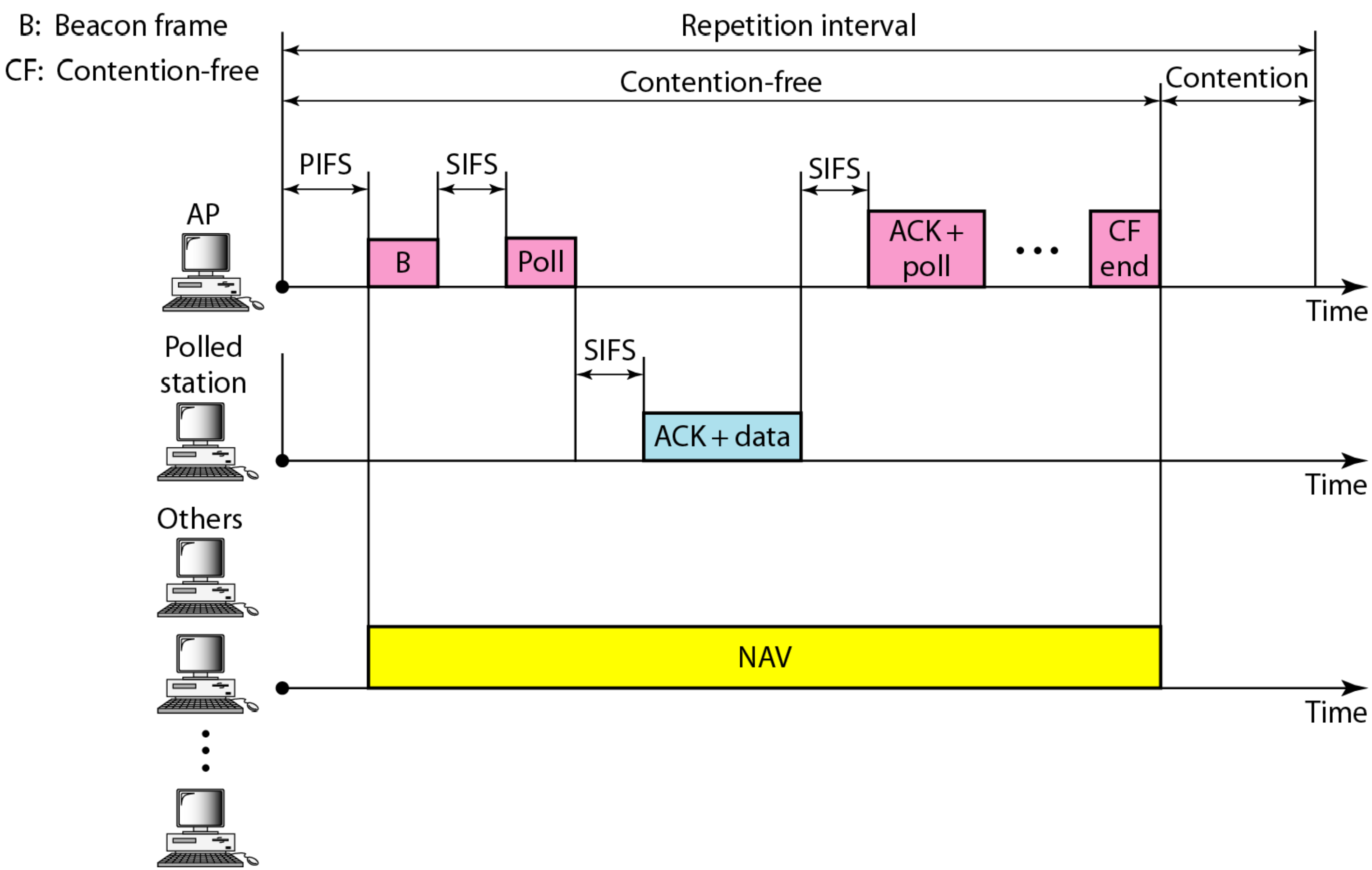


Figure 14.7 Frame format

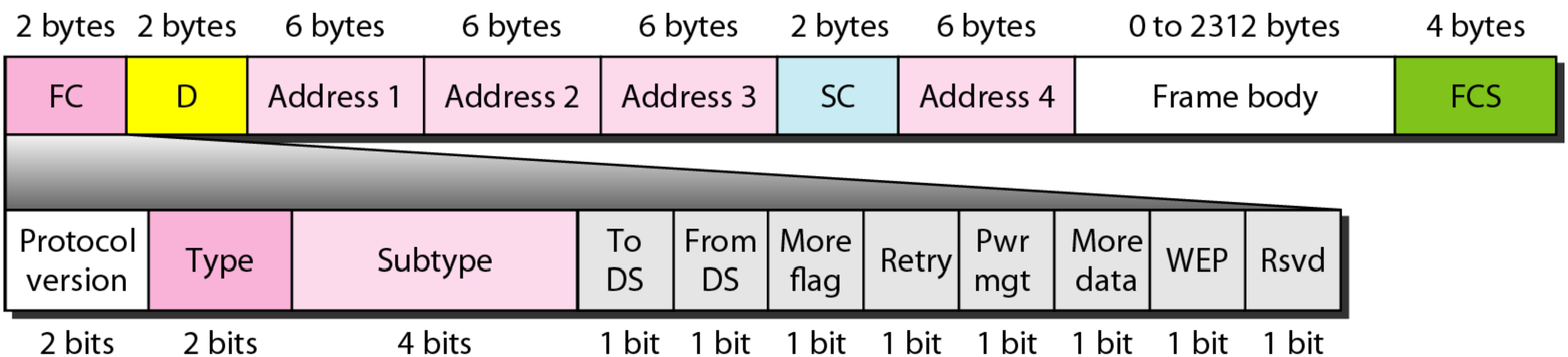


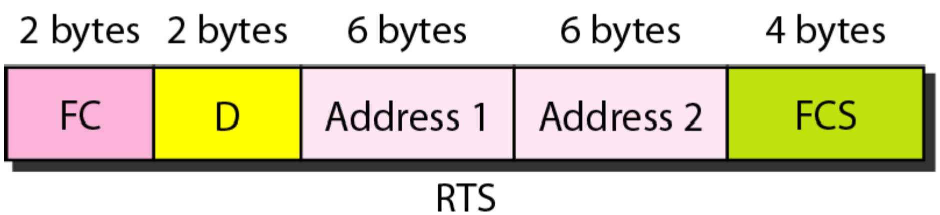
Table 14.1 Subfields

Field	*Explanation*
Version	Current version is 0
Type	Type of information: management (00), control (01), or data (10)
Subtype	Subtype of each type (see Table 14.2)
To DS	Defined later
From DS	Defined later
More flag	When set to 1, means more fragments
Retry	When set to 1, means retransmitted frame
Pwr mgt	When set to 1, means station is in power management mode
More data	When set to 1, means station has more data to send
WEP	Wired equivalent privacy (encryption implemented)
Rsvd	Reserved

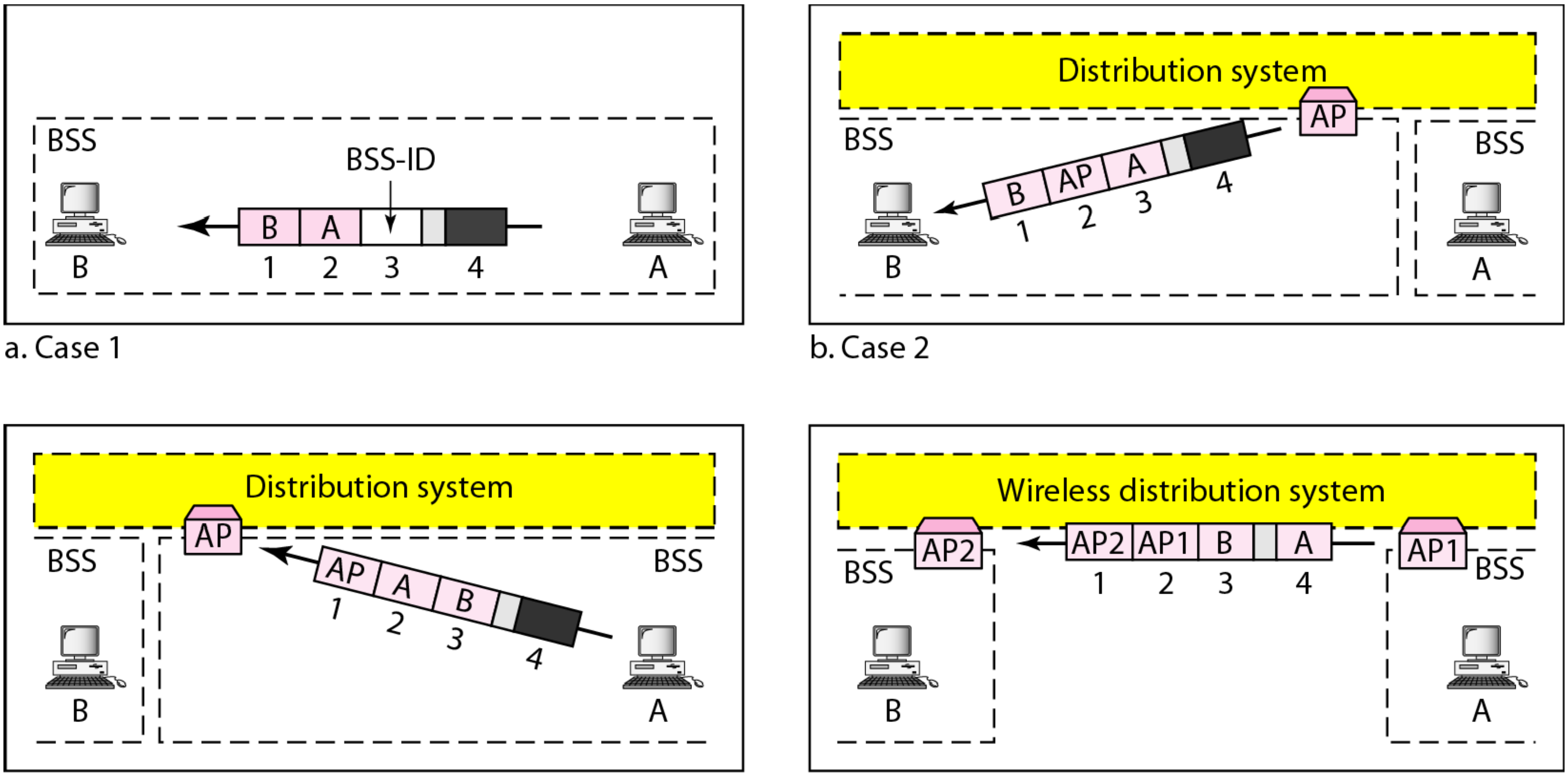
Table 14.2 Values of

Subtype	*Meaning*
1011	Request to send (RTS)
1100	Clear to send (CTS)
1101	Acknowledgment (ACK)

Control frames



Addressing mechanisms



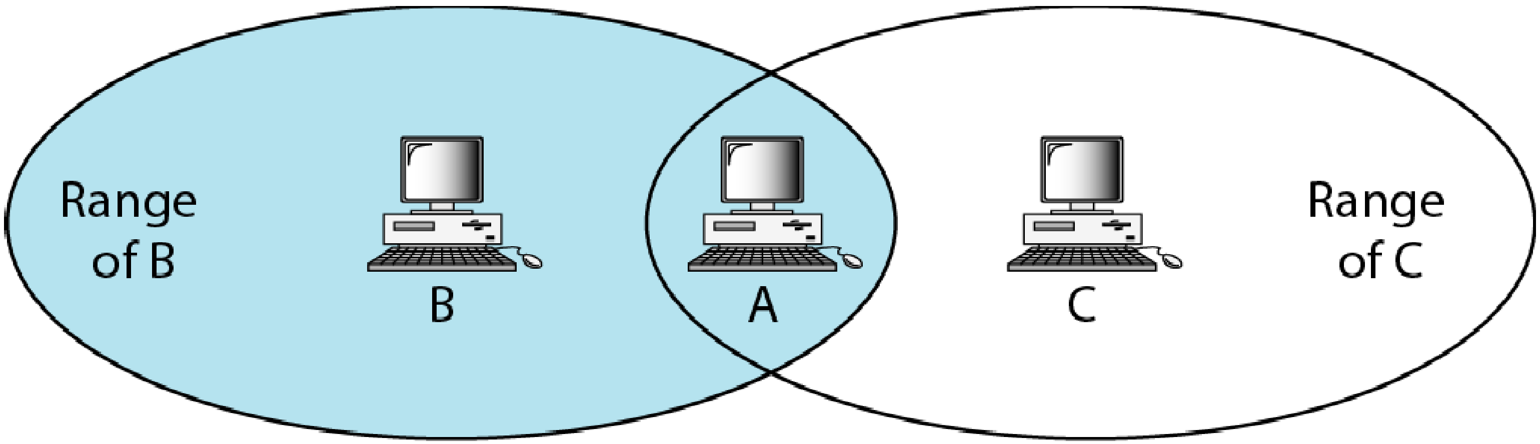
a. Case 1

b. Case 2

c. Case 3

d. Case 4

Hidden station problem

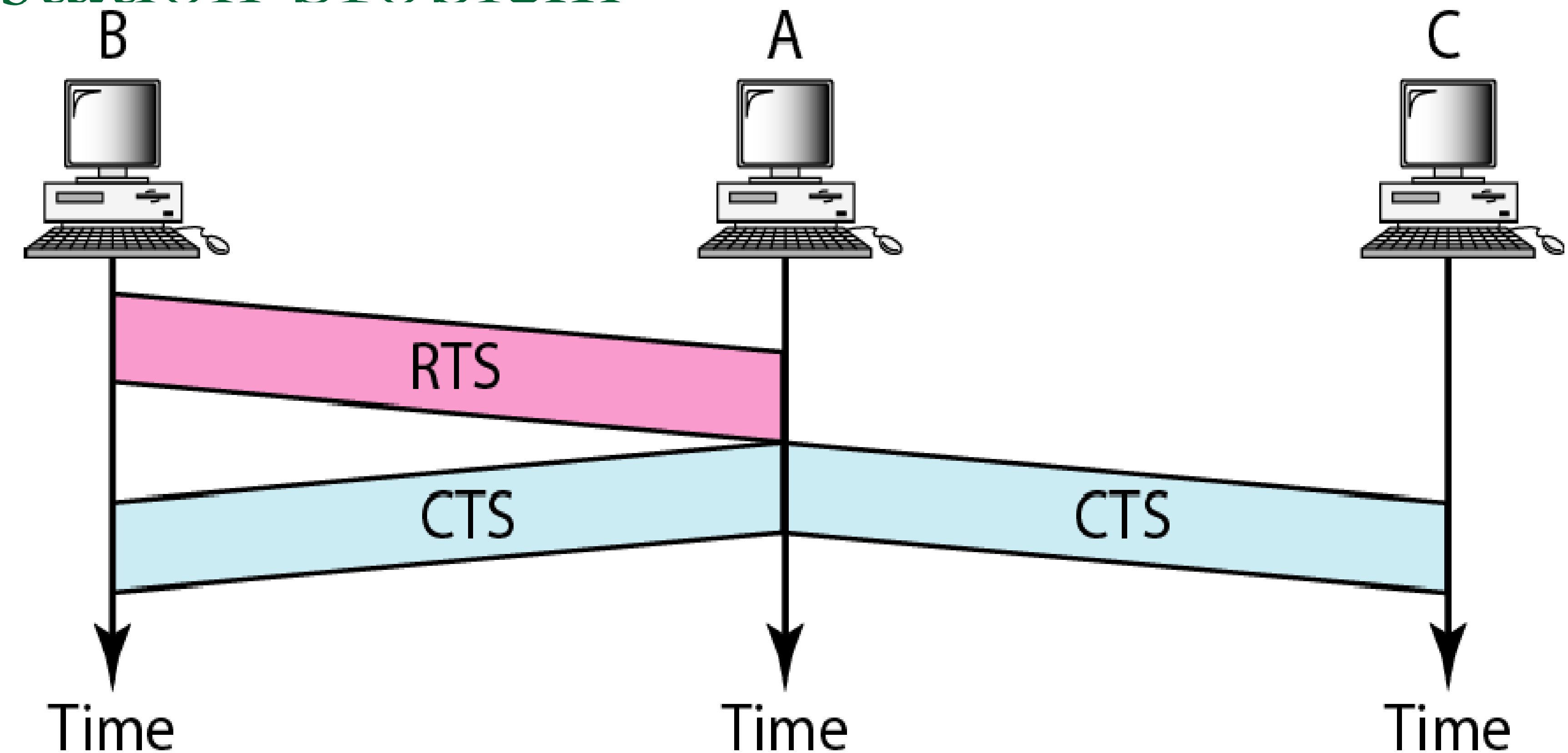


B and C are hidden from each other with respect to A.

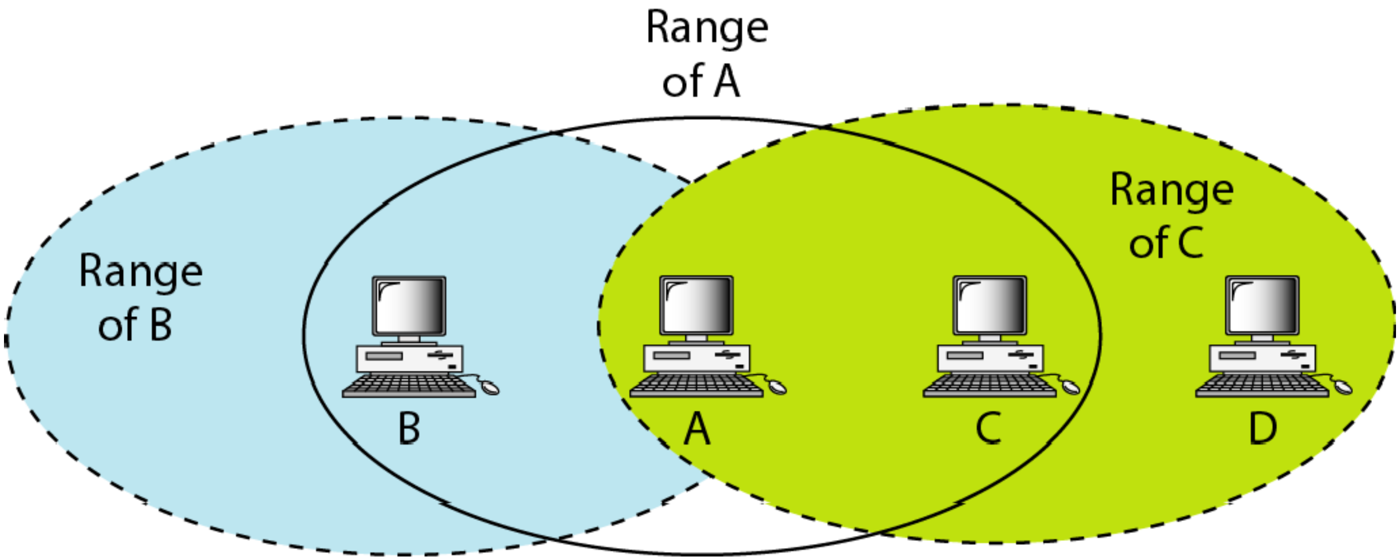
Not

The CTS frame in CSMA/CA handshake can prevent collision from a hidden station.

Use of handshaking to prevent hidden station problem

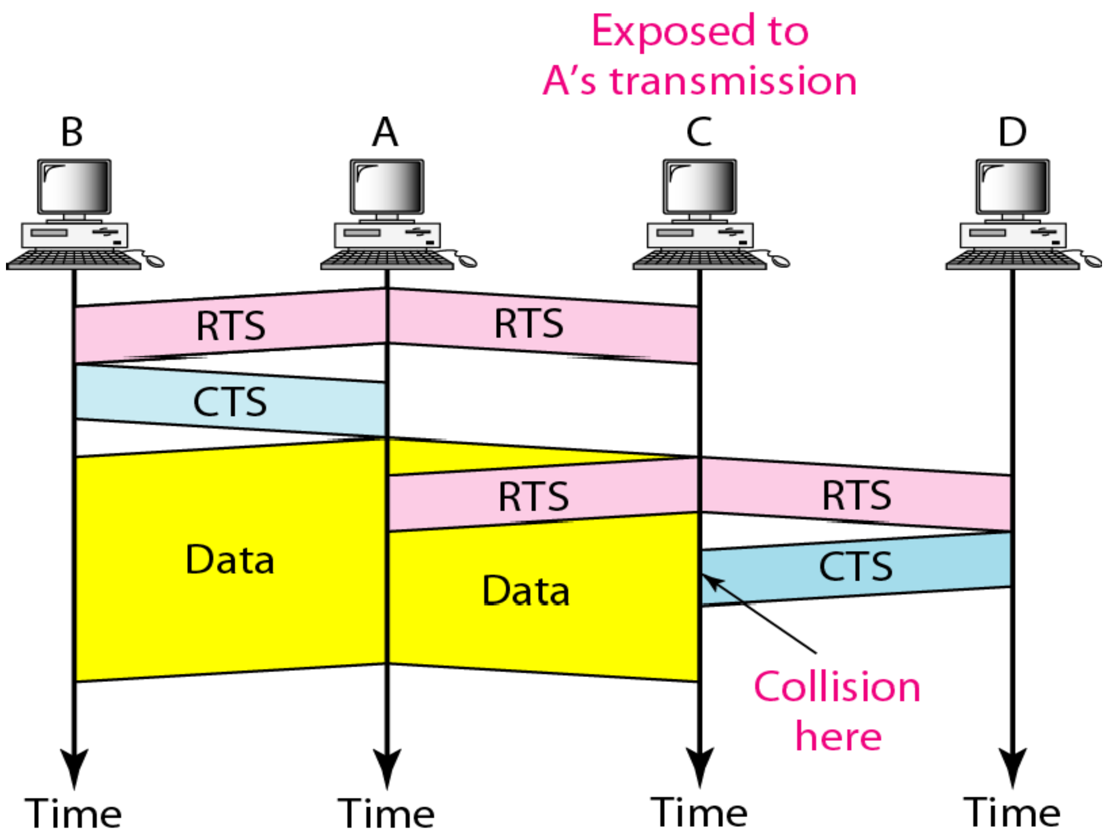


Exposed station problem



C is exposed to transmission from A to B.

Use of handshaking in exposed station problem



Wireless Physical Layer

IEEE 802.11 PHY Transmission Techniques:

1. **FHSS (Frequency Hopping Spread Spectrum)**
   - The signal hops rapidly among many frequency channels within the band.
   - Used in the original **802.11** standard (released in 1997).
   - Pros: Resists interference and eavesdropping.
   - Data rate: Up to 2 Mbps.
   - Less common today.
2. **DSSS (Direct Sequence Spread Spectrum)**
   - Spreads the signal over a wider frequency band by combining it with a pseudo-random noise sequence.
   - Used in **802.11b**.
   - Provides better resistance to interference than FHSS.
   - Data rates: Up to 11 Mbps.

Wireless Physical Layer

- Data rates: Up to 11 Mbps.

3. **Infrared (IR)**
   - Uses infrared light (instead of radio waves) to transmit data.
   - Used in early **802.11** standards.
   - Limited range and requires line-of-sight.
   - Rarely used now.
4. **OFDM (Orthogonal Frequency Division Multiplexing)**
   - Splits the radio signal into multiple smaller sub-signals that are transmitted simultaneously at different frequencies.
   - Used in **802.11a**, **802.11g**, **802.11n**, **802.11ac**, and newer standards.
   - Supports higher data rates (from 54 Mbps up to several Gbps).
   - More efficient and robust against interferen nd multipath effects.

Wireless Physical Layer

- **FHSS** = Quickly switching radio channels to send pieces of the message.
- **DSSS** = Mixing the message with a special code to spread it out over a range of frequencies.
- **Infrared** = Sending the message using light beams.
- **OFDM** = Breaking the message into many smaller parts and sending them all at once on different frequencies.

- **FHSS & DSSS:** Older spreading techniques to reduce interference.
- **Infrared**: Early, line-of-sight technology.
- **OFDM**: Modern, high-speed, robust technique used in most current Wi-Fi standards.

Bridges

- *Means to interconnect individual LANs*
- *Operate at the* ***data link layer***
- *Reasons for bridges:*

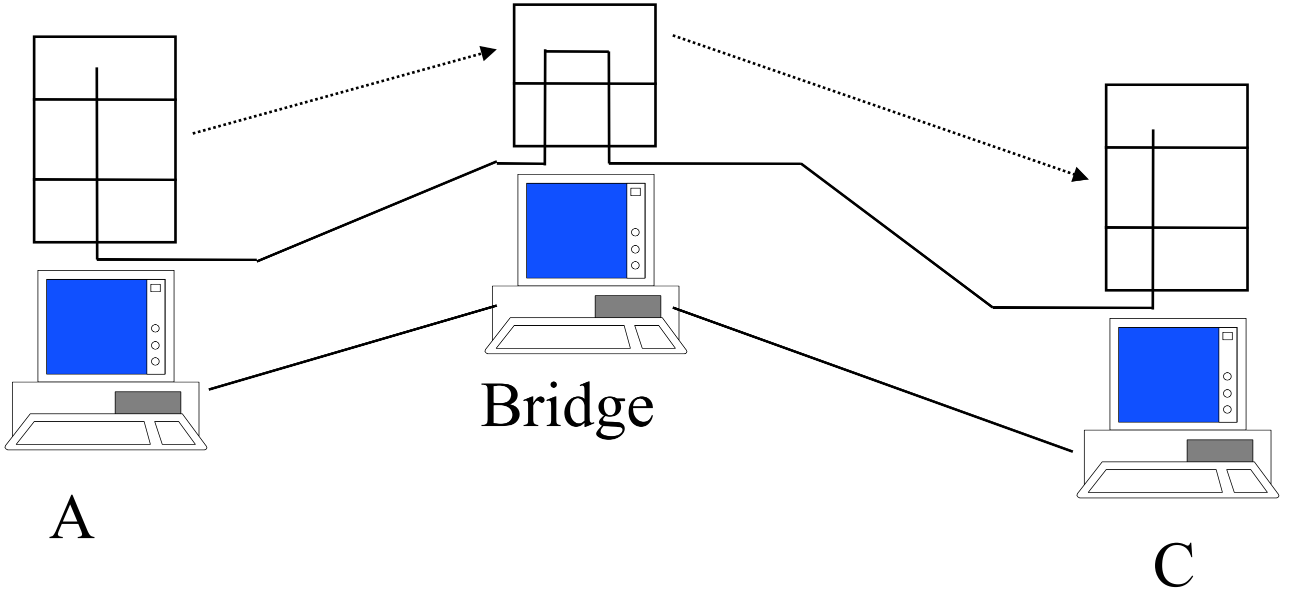
  (1)autonomous divisions (2) different buildings (3) logical grouping (4) physical distance (5) Reliability (6) Security(due to Promiscuous mode)
- *Issues in bridging different LANs:*

  (i) Different frame format (ii) different data rates (iii) Timers at higher layers (iv) different maximum frame lengths

Bridges

- *Multiple LANS can be connected by devices called bridges.*
- *If a bridge is added or detected from the system, reconfiguration of the stations is unnecessary(It means that bridges should be fully transparent).*

Bridges



LANs can be connected by devices called bridges, which operate in the data link layer. Bridges do not examine the network layer header.

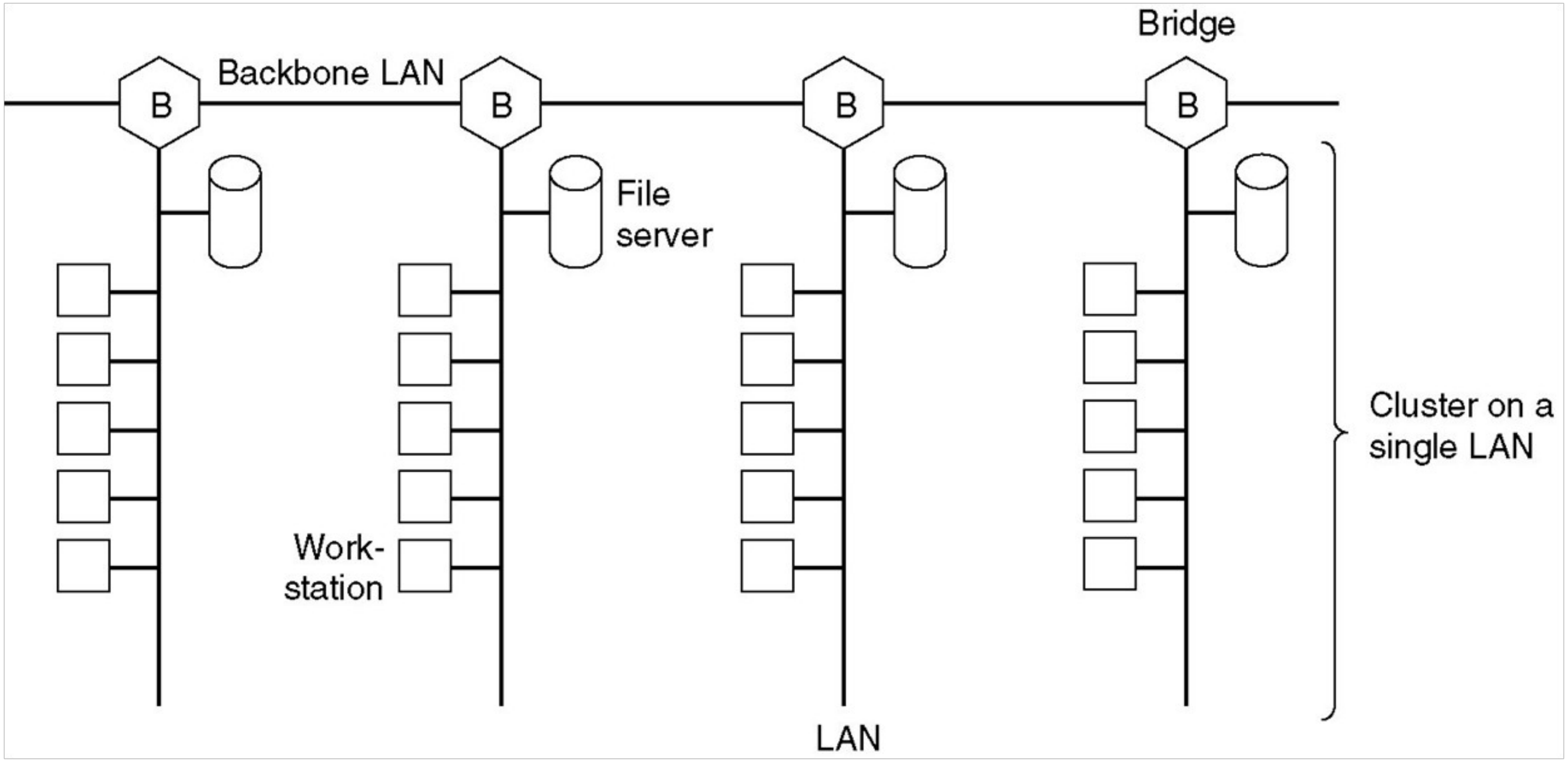
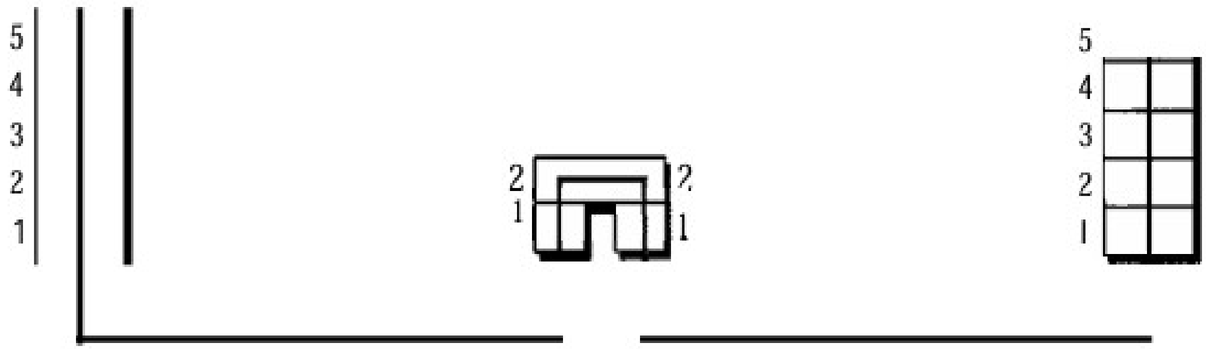


Fig. *Multiple LANs connected by a backbone to handle a total load higher than the capacity of a single LAN*.

Characteristics of Bridges

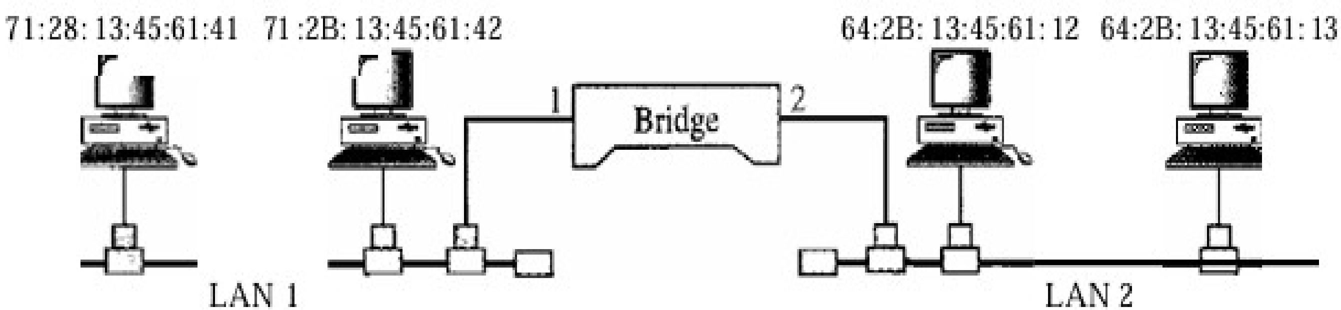
- **Routing Tables**
  - Contains one entry per station of network to which bridge is connected.
  - Is used to determine the network of destination station of a received packet.
- **Filtering**
  - Is used by bridge to allow only those packets destined to the remote network.
  - Packets are filtered with respect to their destination and multicast addresses.
- **Forwarding**
  - the process of passing a packet from one network to another.
- **Learning Algorithm**
  - the process by which the bridge learns how to reach stations on the internetwork.

Figure 15.5 *A bridge connecting two LANs*



Address	Port
71:2B: 13:45:61:41	1
71:2B: 13:45:61:42	1
64:2B: 13:45:61:12	2
64:28:13:45:61:13	2

Bridge Table



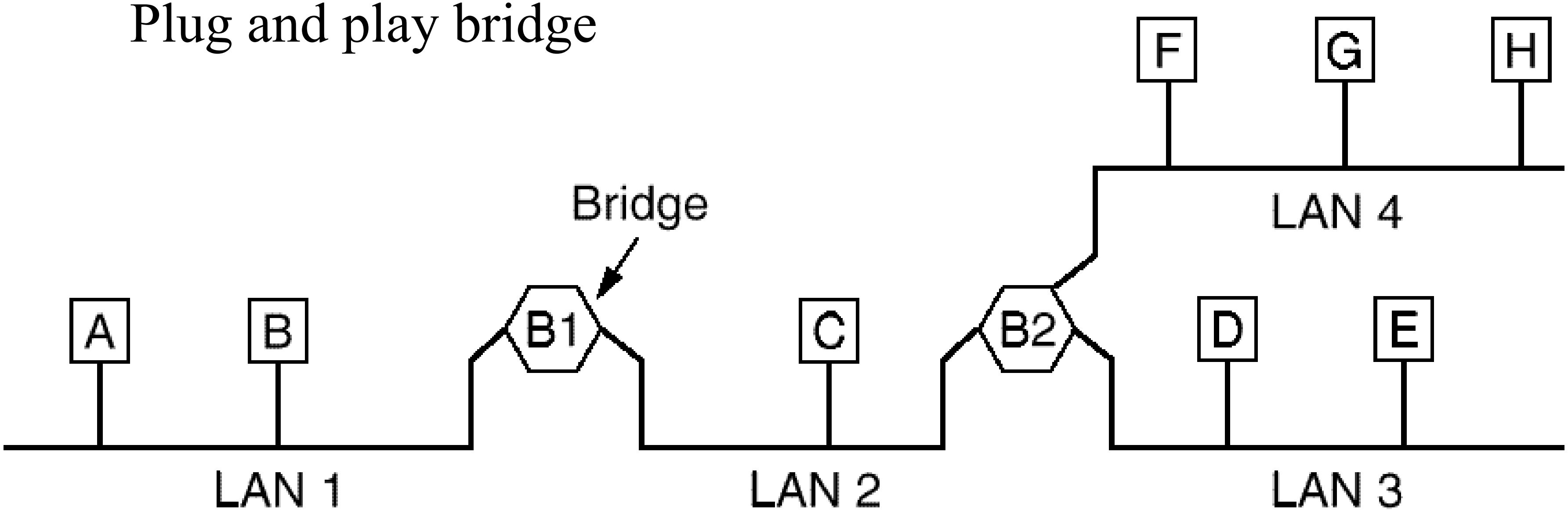
TRANSPARENT BRIDGES

- *When a frame arrives, a bridge must decide whether to discard or forward it.*
- *Each bridge contains hash tables specifying that destination address.*
- *The table can list each possible destination and tell which output line it belongs on.*
- *When the bridges are first plugged in. all the hash tables are empty.*
- *Few days back they are using flooding algorithm for bridges.*
- *Every incoming frame for an unknown destination is output on all the LANs to which the bridge is connected except the one it arrived on.*
- *The algorithm used by the transparent bridges is Backward-learning ,based on source address bridges can tell which machine is accessible on which LAN.*

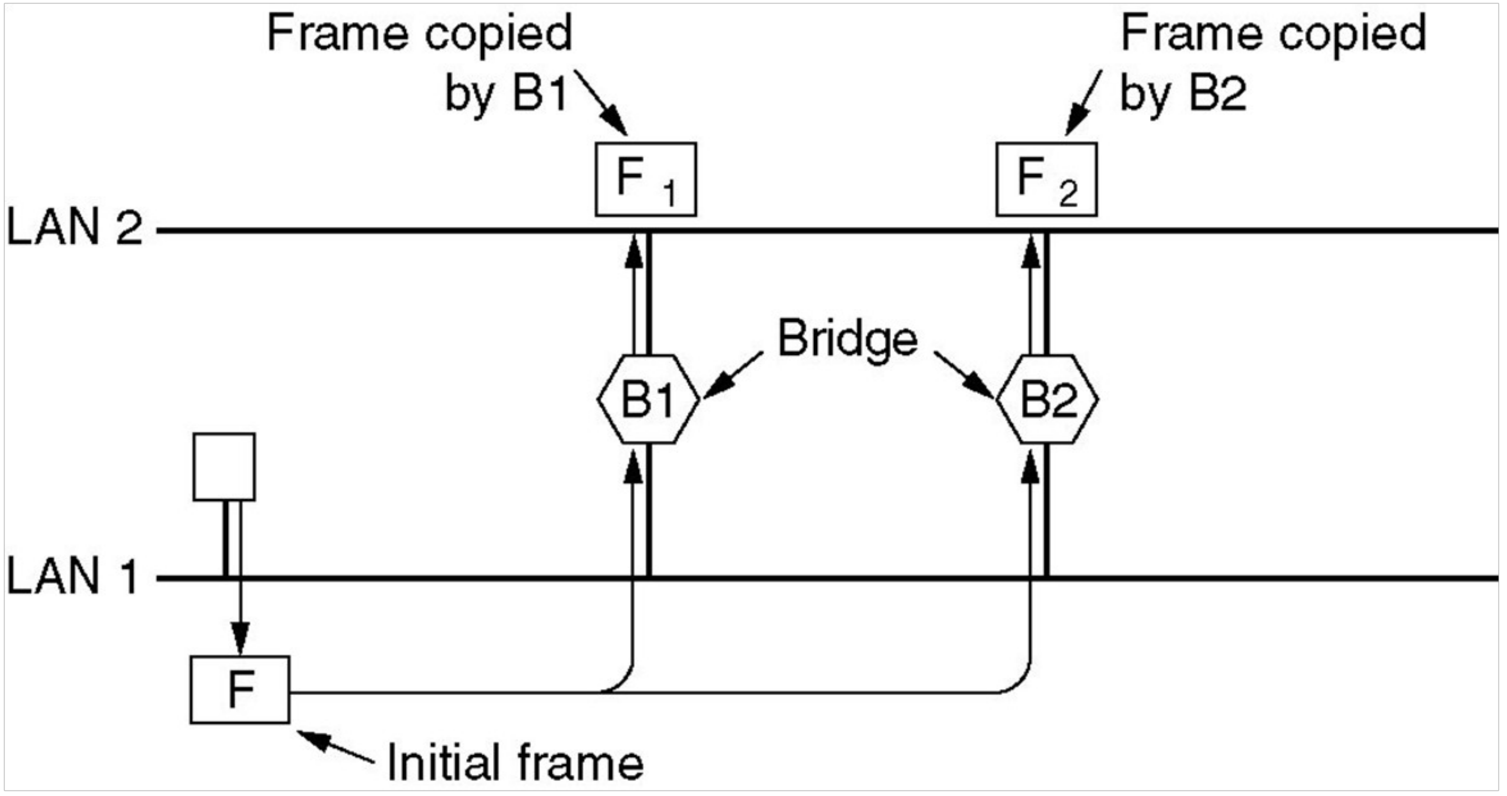
- *3 rules for forwarding the frame.*
- *If destination and source LANs are the same discard the frame.*
- *If destination and source LANs are different , forward the frame.*
- *If the destination LAN is unknown, use flooding.*

Transparent Bridges

Plug and play bridge

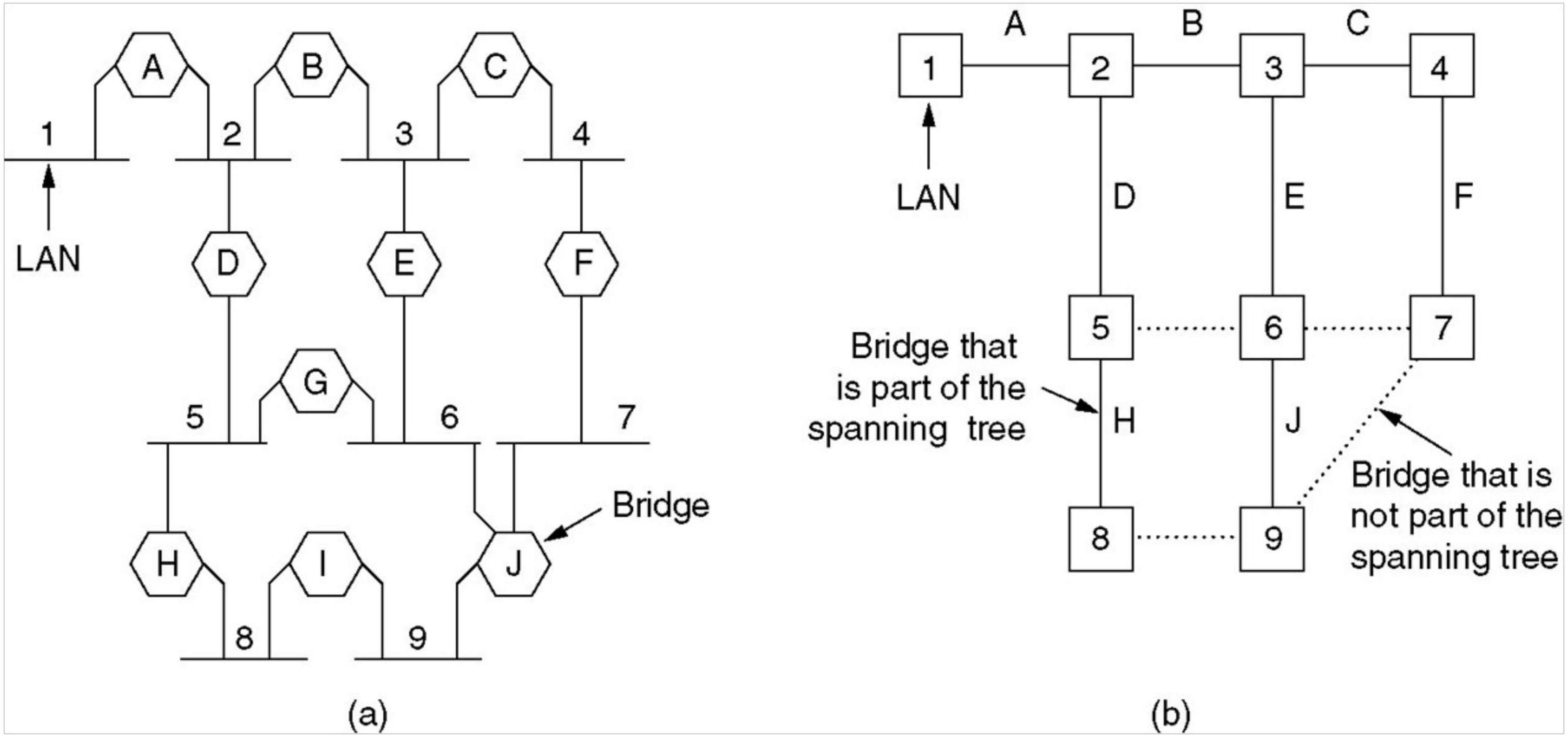


When a frame arrives, a bridge must decide whether to discard or forward it, and if the latter, on which LAN to put the frame. The decision is made by looking up the destination address in a big (hash) table inside the bridge.



Two parallel transparent bridges

SPANNING TREE BRIDGES



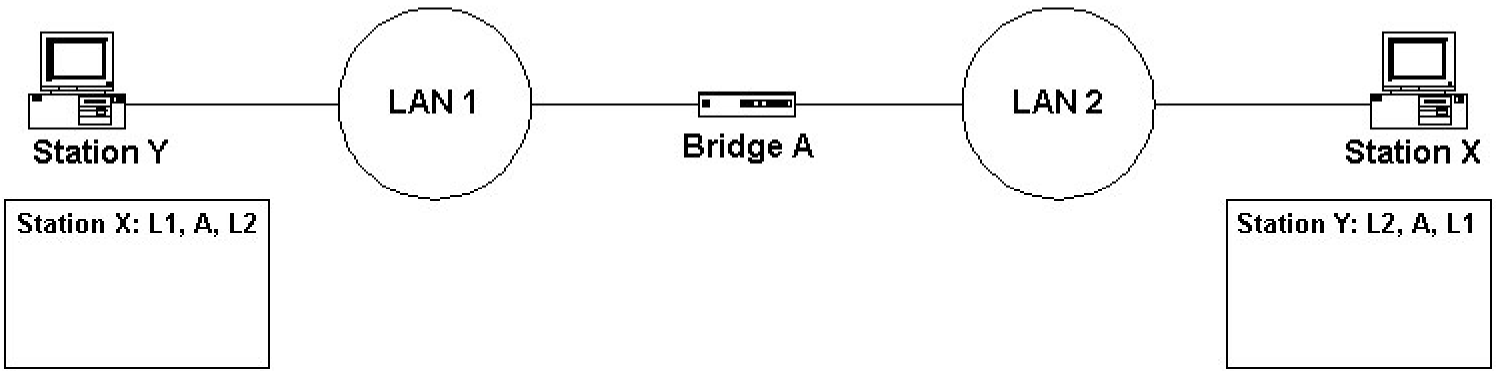
- *(a) Interconnected LANs. (b) A spanning tree covering the LANs. The dotted lines are not part of the spanning tree.*

Spanning Tree Bridges

To build the spanning tree, first the bridges have to choose one bridge to be the root of the tree. They make this choice by having each one broadcast its serial number, installed by the manufacturer, and guaranteed to be unique worldwide. The bridge with the lowest serial number becomes the root. Next, a tree of shortest paths from the root to every bridge and LAN is constructed.

Source routing Bridges

Source routing assumes that the sender of each frame knows whether or not the destination is on its own LAN. When sending a frame to a different LAN, the source machine sets the high-order bit of the destination address to 1, to mark it. Furthermore, it includes in the frame header the exact path that the frame will follow.


Station Y
LAN 1
Bridge A
LAN 2
Station X
Station X: L1, A, L2
Station Y: L2, A, L1

Implicit in the design of source routing bridge is that every machine in the internetwork knows, or can find, the best path to every other machine. How these routes are discovered is an important part of the source routing bridge.

The basic idea is that if a destination is unknown, the source issues a broadcast frame asking where it is. This discovery frame is forwarded by every bridge so that it reaches every LAN on the internetwork. When the reply comes back, the bridges record their identity in it, so that the original sender can see the exact route taken and ultimately choose the best route.

4.4.4 Comparison of 802 Bridges

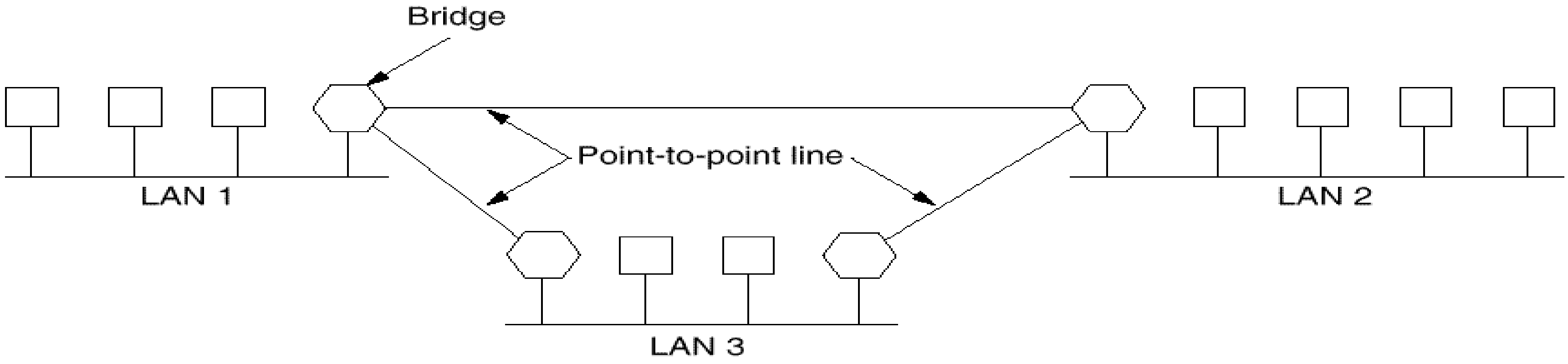
Issue	Transparent bridge	Source routing bridge
Orientation	Connectionless	Connection-oriented
Transparency	Fully transparent	Not transparent
Configuration	Automatic	Manual
Routing	Suboptimal	Optimal
Locating	Backward learning	Discovery frames
Failures	Handled by the bridges	Handled by the hosts
Complexity	In the bridges	In the hosts

Fig. 4-42. Comparison of transparent and source routing bridges.

Remote Bridges

- Used when individual LANs are far apart
- In this case bridges on each LAN are connected via point-to-point links

Remote Bridges



Point to point protocol used between bridges(2 options):

1.Choose some standard point-to-point data link protocol i.e, PPP, putting complete MAC frames in the payload field (best if all LANs are identical)

2. Strip off the MAC header and trailer at source bridge, put what is left in the payload field of PPP. A new MAC header and trailer can be generated at the destination bridge(can not catch errors caused by bad bridge memory)